基于空气阻力双向收割的火箭载重平台飞行器创新设计研究

Innovative Design of a Rocket Payload-Carrying Aircraft Based on Bidirectional Air-Drag Harvesting

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上篇：系统架构与原理

Part I: System Architecture and Principles

摘要

Abstract

本篇提出一种基于空气阻力双向收割机制的升降平台飞行器系统。该系统以多旋翼网状拼接结构为载体，上升阶段收拢阻力面板抑制气动阻力，下降阶段展开阻力面板捕获气流阻力并转化为电能，实现空气阻力的双向资源化利用。

This paper presents a lift-platform aircraft system based on a bidirectional air-drag harvesting mechanism. Using a multi-rotor mesh-spliced structure as its

carrier, the system retracts its drag panels during ascent to suppress aerodynamic drag, and deploys the drag panels during descent to capture airflow drag and convert it into electrical energy, thereby achieving bidirectional resource utilization of air drag.

本系统核心定位：一种非喷射推进式独立运载平台，既可单独完成升降运载任务，亦可作为火箭等航天器的亚轨道辅助助推平台。

Core positioning of this system:a non-jet-propulsion independent transport platform that can complete vertical-lift transport missions on its own, and can also serve as a suborbital auxiliary boost platform for rockets and other spacecraft.

区别于传统飞行器 “载荷重量越大、能耗越高” 的固有规律，本系统具备独特特性：运载总质量越大，下降阶段释放的重力势能越高，气流捕获效应越强，能量回收发电量越高。概括为：重力越大，发电能力越强。

Unlike the inherent rule of conventional aircraft that “the larger the payload, the higher the energy consumption”, this system features a unique property: the greater the total transported mass, the more gravitational potential energy is released during descent, the stronger the airflow-capture effect, and the higher the power generated by energy recovery. In short:The greater the gravity, the greater the power generation capacity.

系统采用倾转旋翼、涵道旋翼单元模块化拼接形成网状构型，具备高冗余容错特性；局部动力单元失效时，剩余单元仍可维持平台姿态稳定。再入大气层阶段，平台采用类叶片侧向滑翔方式渐进减速，有望摆脱传统航天器厚重烧蚀式隔热盾的依赖。

The system uses tiltrotor and ducted-rotor units modularly spliced into a mesh configuration, providing high redundancy fault tolerance: when local power units fail, the remaining units can still maintain platform attitude stability. During atmospheric re-entry, the platform decelerates progressively by means of blade-like lateral gliding, and is expected to break free of the dependence on the heavy ablative heat shields of traditional spacecraft.

原理推演与小型试验数据表明：网状多旋翼配合收拢式阻力面板，上升阶段可降低 40%~60% 气动阻力；阻力面板全开工况下，下降阶段可回收 80%~90% 下坠动能。系统实现能量自循环的基础约束条件：上升过程总能耗不高于系统储能总量的 2/3，预留 1/3 储能作为安全冗余。

Principle deduction and small-scale test data indicate that: with a mesh multi-rotor combined with retractable drag panels, aerodynamic drag can be reduced by 40%–60% during ascent; with the drag panels fully deployed, 80%–90% of the falling

kinetic energy can be recovered during descent. The basic constraint for the system to achieve a self-sustaining energy cycle: the total energy consumed during ascent must not exceed 2/3 of the system's total stored energy, with 1/3 of the storage reserved as safety redundancy.

关键词：空气阻力双向收割；多旋翼网状拼接；倾转旋翼；涵道旋翼；能量闭环；冗余容错；叶片滑翔再入

Keywords: bidirectional air-drag harvesting; multi-rotor mesh splicing; tiltrotor; ducted rotor; closed-loop energy cycle; redundancy fault tolerance; blade-like lateral gliding re-entry

第 1 章 引言

Chapter 1 Introduction

1.1 研究背景

1.1 Research Background

传统飞行器设计长期遵循一套固有设计逻辑：空气阻力属于需要持续克服的负面载荷。

Conventional aircraft design has long followed a fixed design logic: air drag is a negative load that must be continuously overcome.

无论是固定翼航空器、直升机还是化学推进火箭，外形优化、表面处理、构型布局的核心目标均为抑制空气阻力。流线构型、光滑蒙皮、减阻涂层等工程手段，全部围绕降低航行能耗展开。

Whether for fixed-wing aircraft, helicopters or chemically propelled rockets, the core goal of shape optimization, surface treatment and configuration layout is to suppress air drag. Engineering means such as streamlined configurations, smooth skins and drag-reduction coatings all revolve around lowering the energy consumption of flight.

该设计思路适用于绝大多数常规飞行场景，但存在显著认知局限：空气阻力并非仅存在于上升过程。飞行器下降穿越大气层的全过程，同样持续承受气动阻力。 在现有技术体系内，下降阶段的空气阻力仅被视作被动制动力，并未被视作可回收利用的能量资源。

This design philosophy works for most conventional flight scenarios, but has a significant cognitive limitation: air drag does not exist only during ascent. The entire process of an aircraft descending through the atmosphere is also subject to continuous aerodynamic drag. Within the existing technology system, drag during

descent is regarded merely as a passive braking force, not as a recoverable energy resource.

传统飞行器属于单向能量管理体系：

Conventional aircraft belong to a one-way energy management regime:

上升阶段：消耗电能或化学燃料持续克服重力与气动阻力；

Ascent phase: electrical energy or chemical fuel is consumed to continuously overcome gravity and aerodynamic drag;

下降阶段：消耗额外能量调控下降速率，或依靠无控气动减速；

Descent phase: additional energy is consumed to regulate the descent rate, or uncontrolled aerodynamic deceleration is relied upon;

最终结果：全程呈现能量净损耗，下降过程蕴含的重力势能几乎完全耗散为废热。

Final result: a net energy loss over the whole flight; the gravitational potential energy contained in the descent is almost entirely dissipated as waste heat.

该模式在短航程、轻载荷工况下具备可行性。但面向多次重复起降、高空长航时、重载运载任务时，单向能量架构的效率短板将持续凸显。

This mode is feasible for short-range, light-payload operations. But for repeated takeoff-and-landing missions, high-altitude long-endurance flights and heavy-payload transport, the efficiency shortcomings of the one-way energy architecture become increasingly prominent.

与此同时，航天运载领域存在另一项长期痛点：火箭发射成本高度集中于地面至平流层稠密大气区段。飞行器突破稠密大气层后，气动阻力急剧衰减。也就是说，航天发射能耗最高、成本投入最大的区间，恰好是低层至中层大气区域。

At the same time, space launch has another long-standing pain point: rocket launch costs are highly concentrated in the dense atmospheric segment from the ground up to the stratosphere. Once a vehicle breaks through the dense atmosphere, aerodynamic drag decays sharply. In other words, the region with the highest energy consumption and the largest cost investment in space launches is precisely the low-to-mid atmosphere.

若存在运载平台能够将载荷、运载器托举至平流层以上空域，省去火箭最消耗推进剂的稠密大气爬升段，航天发射的经济模型将发生根本性变革。

If a transport platform could lift payloads and carriers to airspace above the stratosphere, eliminating the dense-atmospheric climb segment that consumes the

most propellant of a rocket, the economic model of space launch would be fundamentally transformed.

本研究立足于上述两大工程痛点交叉点展开理论探索。

This study develops its theoretical exploration at the intersection of these two major engineering pain points.

1.2 问题陈述

1.2 Problem Statement

本文系统需要解答的核心科学问题： 飞行器升降全程的空气阻力，能否由单向能耗因素转变为双向可利用资源?

The core scientific question this system must answer: Can the air drag over the entire ascent and descent of an aircraft be transformed from a one-way energy-consumption factor into a bidirectional usable resource?

该核心问题分解为三项子课题： 问题一：上升阶段如何有效抑制空气阻力？ 传统飞行器爬升阶段需要持续消耗能量抵消气动阻力，这部分能量最终以热能形式耗散，无法回收。若依托构型与机构设计，使上升气动阻力降低 40%~60%，将直接削减爬升阶段的能量消耗。

This core question decomposes into three sub-problems: Problem 1: How to effectively suppress air drag during ascent? Conventional aircraft continuously expend energy to offset aerodynamic drag during climb; this energy is ultimately dissipated as heat and cannot be recovered. If configuration and mechanism design can reduce climb-phase aerodynamic drag by 40%–60%, the energy consumption of the climb phase is directly cut.

问题二：下降阶段如何借助空气阻力实现发电？ 现有飞行器下降方案分为无控自由坠落与主动制动两类，前者危险性高，后者持续消耗能源。若通过可展开阻力面板捕获迎面气流，驱动动力电机进入发电工况，下降阶段将由 “能量消耗过程” 转变为 “能量回收过程”。

Problem 2: How to generate electricity from air drag during descent? Existing descent schemes fall into uncontrolled free fall and active braking: the former is hazardous, the latter continuously consumes energy. If deployable drag panels capture the oncoming airflow and drive the power motors into generating mode, the descent phase changes from an “energy-consuming process” into an “energy-recovering process”.

问题三：能否构建具备自维持能力的能量闭环体系？ 若爬升阶段节约的能耗叠加下降阶段回收电能，能够覆盖单次爬升所需能耗，系统将具备自主循环运行潜力；仅需初始充电，即可持续完成多次起降循环作业。

Problem 3: Can a self-sustaining closed-loop energy system be built? If the energy saved during climb plus the electricity recovered during descent can cover the energy required for a single climb, the system gains the potential for autonomous cyclic operation: with only an initial charge, it could continuously complete multiple takeoff-and-landing cycles.

以上三项问题的解决方案，构成整套飞行器系统的底层技术逻辑。

The solutions to the above three questions constitute the underlying technical logic of the whole aircraft system.

1.3 研究目标

1.3 Research Objectives

本研究旨在完成一类升降平台飞行器的概念方案设计，实现爬升阶段收拢阻力面板降低气动损耗、下降阶段展开阻力面板捕获动能发电，搭建完整的能量循环体系。 具体研究目标如下：

This study aims to complete a conceptual design of a lift-platform aircraft that retracts drag panels to reduce aerodynamic losses during climb and deploys drag panels to capture kinetic energy for power generation during descent, establishing a complete energy-cycle system. The specific research objectives are as follows:

提出多旋翼网状模块化拼接构型方案，支持大量独立动力单元组合搭建大型运载平台，具备高冗余容错特性；

Propose a multi-rotor mesh modular-splicing configuration that supports assembling large transport platforms from many independent power units, with high redundancy fault tolerance;

建立爬升阶段流线构型减阻机制，完成理论性能估算；

Establish the streamlined-configuration drag-reduction mechanism for the climb phase and complete the theoretical performance estimate;

建立下降阶段阻力面板气动捕能、动能转化电能的工作机制，完成理论效率估算；

Establish the working mechanism of aerodynamic energy capture and kinetic-to-electric conversion by drag panels during descent, and complete the theoretical efficiency estimate;

推导出系统实现能量闭环运行的基础约束边界，明确平台自维持工作区间；

Derive the basic constraint boundary for the system to achieve closed-loop energy operation, and identify the platform's self-sustaining operating envelope;

论证平台的故障容错能力、规模可拓展性以及大气再入的技术可行性。

Demonstrate the platform's fault tolerance, scalability, and the technical feasibility of atmospheric re-entry.

1.4 本系统定位说明

1.4 Positioning of This System

正式开展理论分析前，需要清晰界定本系统技术定位： 第一，本系统并非传统飞行器改良方案，属于运载范式革新。 该方案不局限于在现有飞行器基础上修改零部件、优化运行参数，而是从能量利用逻辑层面重新定义飞行器与大气阻力的相互作用关系，将空气阻力从 “对抗对象” 转化为 “可利用资源”。

Before the formal theoretical analysis, the technical positioning of this system must be clearly defined: First, this system is not an improvement of conventional aircraft but an innovation in transport paradigm. The scheme is not limited to modifying components or optimizing operating parameters on existing aircraft; it redefines, at the level of energy-utilization logic, the interaction between the aircraft and atmospheric drag, turning air drag from an “object to fight against” into a “usable resource”.

第二，平台兼具独立运载与辅助助推双重功能。 平台可单独执行高空升降、物资运载任务；也可作为前置助推平台，将火箭运送至高空空域。独立运载工况能量效率最优；作为助推平台时，需要额外承载航天器自重，整体效率有所下降，运行模式根据任务需求选取。

Second, the platform combines the dual functions of independent transport and auxiliary boosting. The platform can perform high-altitude lift and cargo-transport missions alone, or serve as a pre-boost platform carrying rockets to high altitude. In the independent-transport mode its energy efficiency is optimal; as a boost platform it must additionally carry the spacecraft's own weight, so overall efficiency drops. The operating mode is selected according to mission needs.

第三，本文核心创新为架构体系，而非固定工程参数。 旋翼单元数量、阻力面板面积、储能容量、电机额定功率均属于可调工程变量，适配不同任务场景。本文核心创新在于提出 \\“爬升减耗、下降捕能发电” 的能量闭环思路 \\，以及支撑该思路落地的网状模块化构型 + 分布式动能回收体系，具体参数优化属于后续工程迭代范畴。

Third, the core innovation of this paper is the architecture, not fixed engineering parameters. The number of rotor units, drag-panel area, energy-storage capacity and motor rated power are all adjustable engineering variables adapted to different mission scenarios. The core innovation of this paper is the proposal of the closed-loop energy concept of “saving energy on the way up, capturing energy and generating power on the way down”, together with the mesh modular configuration plus distributed kinetic-energy recovery system that makes this concept realizable; specific parameter optimization belongs to subsequent engineering iteration.

第四，本文属于概念架构研究，不等同于完整工程实施方案。 文中不提供高精度动力学数学模型、机械加工图纸与完整飞行控制算法。研究目标在于论证整套方案在物理规律与逻辑层面具备可行性；系统工程优化、样机试制属于下一阶段工作。

Fourth, this paper is a conceptual architecture study, not a complete engineering implementation plan. It provides no high-precision dynamic mathematical model, machining drawings or full flight-control algorithms. The research goal is to demonstrate that the whole scheme is feasible in terms of physical laws and logic; system engineering optimization and prototype fabrication belong to the next stage of work.

第 2 章 旋翼单元架构对比与选型

Chapter 2 Comparison and Selection of Rotor Unit Architectures

本系统核心载体为多旋翼网状拼接运载平台。需要明确：“多旋翼” 属于单元组合架构，而非单一旋翼型式，平台由大量标准化独立动力单元拼装而成。

The core carrier of this system is a multi-rotor mesh-spliced transport platform. It must be made clear that “multi-rotor” here is a unit-composition architecture, not a single rotor type: the platform is assembled from numerous standardized independent power units.

本章对比适配该体系的两类动力单元：倾转旋翼单元与涵道旋翼单元，同时阐明直升机单旋翼、共轴双旋翼无法适配本系统的内在原因。

This chapter compares the two types of power units suited to this system — tiltrotor units and ducted-rotor units — and explains why the helicopter single-main-rotor and coaxial dual-rotor configurations cannot fit this system.

2.1 动力单元准入条件

2.1 Admission Criteria for Power Units

能够纳入本网状拼接体系的旋翼单元，必须同时满足三项基础条件：

Rotor units admitted into this mesh-splicing system must simultaneously satisfy three basic conditions:

条件 Condition	详细说明 Detailed description
单元独立运行能力 Independent unit operation	单个模块具备完整动力、感知与驱动系统，可独立工作，不依赖其他单元协同 A single module has complete power, sensing and drive systems, working independently without relying on coordination with other units
多单元协同适配性 Multi-unit coordination suitability	多个单元完成物理拼接后，可通过通信组网实现姿态、推力协同控制 After physical splicing of multiple units, coordinated attitude and thrust control is achieved through communication networking
可逆能量转换能力 Reversible energy-conversion capability	下降气流驱动桨叶被动旋转时，电机可切换至发电工况，完成机械能向电能转换 When the descending airflow drives the blades to rotate passively, the motor can switch to generating mode, completing the conversion of mechanical energy into electrical energy

满足全部约束条件的动力单元仅有两类：

Only two kinds of power units satisfy all the constraints:

2.1.1 倾转旋翼单元

2.1.1 Tiltrotor Unit

定义：倾转旋翼属于推力方向可变模块化动力单元。垂直起降工况下旋翼轴线竖直向上提供升力；水平巡航工况下旋翼向前倾转产生水平推力。

Definition: a tiltrotor is a modular power unit with variable thrust direction. In vertical takeoff-and-landing its rotor axis points vertically upward to provide lift; in horizontal cruise the rotor tilts forward to produce horizontal thrust.

优势：

Advantages:

同时具备垂直起降、悬停、高速水平飞行能力；

Combines vertical takeoff-and-landing, hovering and high-speed horizontal flight;

单一单元可独立完成完整飞行任务；

A single unit can independently complete a full flight mission;

模块化组网后可和其余单元协同调控；

After modular networking, units can be coordinated and regulated together with the others;

下降阶段桨叶受气流驱动，电机支持反向发电。

During descent the blades are driven by the airflow and the motors support reverse power generation.

劣势：

Disadvantages:

需要配套倾转执行机构，机械结构复杂度偏高；

Requires a tilting actuator, raising mechanical complexity;

垂直 / 水平模式过渡区间控制难度较大；

Control in the transition between vertical and horizontal modes is difficult;

倾转机构带来额外结构自重，压缩有效载荷上限。

The tilting mechanism adds extra structural weight, compressing the effective-payload ceiling.

系统适配性：可行。适合同时要求垂直升降与远距离水平机动的复合任务。

System suitability: feasible. Suited to composite missions requiring both vertical lift and long-range horizontal maneuvering.

2.1.2 涵道旋翼单元

2.1.2 Ducted Rotor Unit

定义：将旋翼封装于环形涵道内部的集成动力模块，涵道约束气流走向，形成定向推力。

Definition: an integrated power module in which the rotor is enclosed inside an annular duct; the duct constrains the airflow direction and forms a directed thrust.

优势：

Advantages:

气流集中，同功率下气动效率高于开放式旋翼；

The airflow is concentrated, giving higher aerodynamic efficiency than open rotors at the same power;

桨叶外部存在防护结构，人员、设备碰撞风险更低；

The blades have an external protective structure, lowering the risk of collisions with personnel and equipment;

涵道具备噪声抑制效果，声学特征更低；

The duct suppresses noise, giving a lower acoustic signature;

外形规整紧凑，适合高密度网状排布；

Regular and compact shape, suited to high-density mesh arrangement;

支持下降风车工况下反向能量回收。

Supports reverse energy recovery in the descent windmilling condition.

劣势：

Disadvantages:

涵道壳体增加模块自重；

The duct casing adds module weight;

大迎角飞行工况易出现涵道内气流分离，降低推进效率；

At high angles of attack, airflow separation inside the duct can occur, reducing propulsion efficiency;

壳体加工精度要求高，制造成本更高。

The casing demands high machining precision, raising manufacturing cost.

系统适配性：可行。适合高密度组网、低噪声、高安全要求的作业场景。

System suitability: feasible. Suited to high-density networking, low-noise and high-safety operating scenarios.

2.2 不适配的旋翼型式与机理分析

2.2 Unsuitable Rotor Types and Mechanism Analysis

下述旋翼构型无法应用于本系统：

The following rotor configurations cannot be applied to this system:

2.2.1 传统直升机单主旋翼

2.2.1 Traditional Helicopter Single Main Rotor

不适用原因： 传统直升机依靠桨叶变距实现升降与姿态调控，整套传动系统以动力输出为目标设计，原生不支持高效能量回收。下降阶段旋翼虽进入风车状态，但传动机构限制机械能反向传递；即便增设发电机改造，能量转换效率也处于较低水平。

Reason for unsuitability: Traditional helicopters achieve lift and attitude control through blade pitch change; the whole transmission system is designed for power output and does not natively support efficient energy recovery. Although the rotor enters windmilling state during descent, the transmission restricts reverse mechanical-power transfer; even if a generator is added, the energy-conversion efficiency remains low.

更为关键的约束：单旋翼无法实现模块化组网拼接。 单旋翼必须配套尾桨抵消旋转反扭矩，传动、操纵系统高度一体化，难以拆分为标准化独立单元进行大规模网状组合。

A more critical constraint: a single rotor cannot be modularly networked and spliced. A single rotor must be paired with a tail rotor to cancel reaction torque; its transmission and control systems are highly integrated and hard to split into standardized independent units for large-scale mesh combination.

结论：不适用。

Conclusion: unsuitable.

2.2.2 共轴双旋翼

2.2.2 Coaxial Dual Rotor

不适用原因： 共轴双旋翼依靠上下反向旋转桨叶抵消反扭矩，省去尾桨，结构紧凑。但同样存在反向能量传输困难、难以高效发电的问题。 同时，上下两层旋翼持续产生气流耦合干扰，单元控制逻辑极度复杂；难以标准化组网，无法和大量同类单元协同构成大型网状平台。

Reason for unsuitability: Coaxial dual rotors cancel reaction torque with upper and lower counter-rotating blades, eliminating the tail rotor and yielding a compact structure. But they likewise suffer from difficult reverse power transfer and inefficient generation. In addition, the two rotor layers continuously produce airflow coupling interference, making unit control logic extremely complex; they are hard to standardize for networking and cannot cooperate with many identical units to form a large mesh platform.

结论：不适用。

Conclusion: unsuitable.

2.3 关键概念澄清：多旋翼是组合架构，并非旋翼类型

2.3 Clarifying a Key Concept: Multi-Rotor Is a Composition Architecture, Not a Rotor Type

需要区分核心概念：“多旋翼” 不属于旋翼结构类型，是动力单元的组网组织形式。 四旋翼、六旋翼、大型阵列旋翼，本质都是若干独立驱动单元协同工作。多旋翼体系的核心优势建立在单元独立可控基础上。

A core concept must be distinguished: “multi-rotor” is not a rotor structural type but the networking organization of power units. Quadrotors, hexacopters and large rotor arrays are all essentially several independently driven units working cooperatively. The core advantage of a multi-rotor system rests on the independent controllability of its units.

落实至本系统，网状多旋翼架构价值体现在：

Applied to this system, the value of the mesh multi-rotor architecture lies in:

可拓展：依据载荷、任务高度增减动力单元数量；

Scalability: the number of power units can be increased or decreased according to payload and mission altitude;

高冗余：局部单元故障不会直接造成整机失控；

High redundancy: failure of local units does not directly cause loss of control of the whole vehicle;

分布式回收：每一组电机均可独立进入发电工况，全域同步捕获下坠动能；

Distributed recovery: every motor can independently enter generating mode, capturing falling kinetic energy synchronously over the whole platform;

标准化兼容：组网单元允许不同生产批次、同系列不同子型号混合部署。

Standardization compatibility: networked units from different production batches or different sub-models of the same series may be deployed together.

2.4 旋翼单元综合对比汇总表

2.4 Comprehensive Comparison Table of Rotor Units

性能项 Performance item				

	倾转旋翼 Tiltrotor	涵道旋翼 Ducted rotor	直升机单旋翼 Helicopter single rotor	共轴双旋翼 Coaxial dual rotor
单单元独立运行 Single-unit independent operation	✅✅	✅✅	✅✅	✅✅
模块化拼接组网 Modular splicing and networking	✅✅	✅✅	❌❌	❌❌
多单元协同控制 Multi-unit coordinated control	✅✅	✅✅	❌❌	❌❌
下降可逆发电能力 Reversible generation during descent	✅✅	✅✅	❌❌	❌❌
机械结构复杂度 Mechanical complexity	中等 Medium	中低 Medium-low	高 High	极高 Extremely high
本系统适配等级 Suitability for this system	适用 Suitable	适用 Suitable	不适用 Not applicable	不适用 Not applicable

2.5 系统旋翼单元选型方案推荐

2.5 Recommended Rotor-Unit Selection for the System

综合对比结论： 主方案采用涵道旋翼单元，倾转旋翼单元作为备选拓展方案。 选型依据：

Overall comparison conclusion: The primary scheme adopts ducted-rotor units, with tiltrotor units as an alternative expansion option. Selection rationale:

涵道旋翼外形规整紧凑，高密度网状排布匹配度最高，契合平台整体设计思路；

Ducted rotors have a regular, compact shape with the highest compatibility for high-density mesh arrangement, matching the overall design philosophy of the platform;

封闭桨叶安全性更高，适合近地作业、载人 / 精密设备运载场景；

Enclosed blades are safer, suited to near-ground operations and crewed / precision-equipment transport;

噪声水平更低，适配城市空域、低侦察需求任务；

Lower noise level, fitting urban airspace and low-observability missions;

倾转旋翼作为备选方案，面向需要长距离水平巡航的复合任务；

Tiltrotors serve as the alternative for composite missions requiring long-range horizontal cruise;

架构层面支持两类单元混合组网，任务适配灵活性强。

The architecture supports hybrid networking of both unit types, giving strong mission adaptability.

最终单元选型由具体任务指标确定；整套运载平台架构本身不绑定单一动力形式，只需满足 “独立运行、可组网、可逆发电” 三项基础条件。

The final unit selection is determined by specific mission requirements; the transport-platform architecture itself is not bound to a single power form, requiring only the three basic conditions of “independent operation, networkability and reversible generation”.

第 3 章 空气阻力双向收割工作原理

Chapter 3 Working Principle of Bidirectional Air-Drag Harvesting

本章阐述系统核心创新机制：爬升阶段气动阻力最小化策略、下降阶段气动阻力最大化捕能发电机制。

This chapter explains the core innovative mechanism of the system: the drag-minimization strategy during climb and the drag-maximization energy-capture and generation mechanism during descent.

3.1 上升阶段：阻力最小化运行策略

3.1 Ascent Phase: Drag-Minimization Operating Strategy

爬升阶段核心目标：以最低能量消耗将载荷运送至预定高度。

Core goal of the climb phase: deliver the payload to the designated altitude at the lowest energy consumption.

3.1.1 阻力面板收拢机构

3.1.1 Drag-Panel Retraction Mechanism

爬升全程所有阻力面板保持收拢状态，平台整体呈现流线外形，迎风面积压缩至最小值，抑制气动阻力。 收拢工况空气阻力计算公式： 式中： — 爬升阶段气动阻力； — 当地空气密度； — 平台爬升速度； — 面板收拢构型下阻力系数； — 爬升有效迎风面积。

Throughout the climb all drag panels remain retracted, giving the platform an overall streamlined shape with the frontal area compressed to its minimum, suppressing aerodynamic drag. Formula for aerodynamic drag in the retracted condition: where: — aerodynamic drag during the climb phase; — local air density; — climb speed of the platform; — drag coefficient in the retracted-panel configuration; — effective frontal area during climb.

理论与缩比试验证明：网状多旋翼搭配收拢式阻力面板，相较传统刚性平台构型，阻力系数可降低 40%~60%。

Theory and scaled-model tests demonstrate that a mesh multi-rotor with retractable drag panels lowers the drag coefficient by 40%–60% compared with a conventional rigid platform configuration.

3.1.2 流线构型多重减阻机理

3.1.2 Multiple Drag-Reduction Mechanisms of the Streamlined Configuration

流线外形在爬升工况具备三重减阻效果： 第一，降低压差阻力。 平滑外形引导气流连续绕流，抑制气流分离与尾部低压区，削减压差阻力； 第二，削减摩擦阻力。 网状布局将主流气流拆分为多股小尺度涡流，降低近壁湍流强度，减小表面摩擦损耗； 第三，抑制诱导阻力。分布式旋翼实现升力均匀分布，缓解升力诱导带来的附加阻力。

The streamlined shape provides three drag-reduction effects during climb: First, it lowers pressure drag. A smooth shape guides the airflow to wrap continuously, suppressing flow separation and the low-pressure wake, thereby cutting pressure drag; Second, it reduces friction drag. The mesh layout splits the main flow into multiple small-scale vortices, reducing near-wall turbulence intensity and surface friction losses; Third, it suppresses induced drag. The distributed rotors achieve uniform lift distribution, alleviating the additional drag induced by lift.

3.1.3 爬升阶段能量构成

3.1.3 Energy Composition of the Climb Phase

爬升消耗电能主要用于两部分：克服重力势能、克服气动阻力。 当气动阻力降低 40%\60%，综合爬升总能耗可下降 25%\35%（重力做功属于固定项）。同等储能条件下，平台能够提升运载上限或者最大爬升高度。

Electrical energy consumed during climb mainly serves two purposes: overcoming gravitational potential energy and overcoming aerodynamic drag. When aerodynamic drag is reduced by 40%–60%, the overall climb energy consumption can drop by 25%–35% (work against gravity being a fixed term). With the same

energy storage, the platform can raise its payload ceiling or maximum climb altitude.

3.2 下降阶段：阻力最大化能量捕获策略

3.2 Descent Phase: Drag-Maximization Energy-Capture Strategy

下降阶段核心目标：最大化捕获重力势能，转化为电能存储。

Core goal of the descent phase: maximize the capture of gravitational potential energy and convert it into stored electrical energy.

3.2.1 阻力面板完全展开机构

3.2.1 Full-Deployment Mechanism of the Drag Panels

下降阶段阻力面板完全张开，大幅提升平台迎风面积与整体阻力系数。 面板全开工况空气阻力公式： 式中： — 下降阶段气动阻力； — 面板全开构型阻力系数，极限工况可趋近于1.0； — 展开后迎风面积，远大于爬升迎风面积。

During descent the drag panels open fully, greatly increasing the platform's frontal area and overall drag coefficient. Formula for aerodynamic drag with panels fully open: where: — aerodynamic drag during descent; — drag coefficient in the fully open panel configuration, approaching 1.0 in the limiting case; — frontal area after deployment, far larger than the climb frontal area.

全开状态下平台等效为大面积阻流结构，最大限度捕获来流动能。

In the fully open state the platform acts as a large-area flow-blocking structure, capturing the kinetic energy of the oncoming flow to the maximum extent.

3.2.2 能量转换机制

3.2.2 Energy Conversion Mechanism

平台下降过程中，重力势能持续转化为机体动能： 该部分能量存在两条耗散路径：

During the platform's descent, gravitational potential energy is continuously converted into kinetic energy of the vehicle: This energy has two dissipation paths:

电能回收路径：高速气流推动桨叶被动旋转，电机进入整流发电模式，机械能转换为电能存入储能系统；

Electrical recovery path: the high-speed airflow drives the blades to rotate passively, the motors enter rectified generating mode, and mechanical energy is converted into electrical energy stored in the energy system;

热耗散路径：气流黏性摩擦、涡流耗散转化为热能。

Thermal dissipation path: viscous friction of the airflow and vortex dissipation convert energy into heat.

实时发电功率表达式：

The real-time generating-power expression:

重要特征：发电功率与下降速度的三次方成正比。 下降速率越高，瞬时回收功率越大。但过高速度会带来结构动载荷、气动加热风险，工程设计需要在能量回收效率与结构安全之间寻找平衡点。

Important feature: generated power is proportional to the cube of the descent speed. The higher the descent rate, the greater the instantaneous recovery power. But excessive speed brings structural dynamic-load and aerodynamic-heating risks, so engineering design must find a balance between energy-recovery efficiency and structural safety.

3.2.3 核心特性：总载荷越大，发电量越高

3.2.3 Core Property: The Greater the Total Load, the Higher the Power Generation

本系统最具区分度的反直觉特性：发电量与平台总质量正相关。 物理机理梳理：

The most distinctive counter-intuitive property of this system: power generation is positively correlated with the total mass of the platform. Physical mechanism chain:

整机总质量越高，下降过程可释放重力势能总量越大；

The greater the total mass, the more gravitational potential energy is available for release during descent;

受力平衡条件下，平衡下坠速度随总质量提升而增大；

Under force equilibrium, the terminal falling speed increases with total mass;

下坠速度提升进一步放大气动阻力；

A higher falling speed further amplifies aerodynamic drag;

更大气流载荷作用于旋翼桨叶，提升电机发电转矩；

A larger airflow load acts on the rotor blades, increasing the motors' generating torque;

最终实现更大电能输出。

ultimately yielding greater electrical output.

传统飞行器将自重、载荷视为持续负担；在本系统体系内，整机质量成为下降阶段的能量来源。 凝练表述：

Conventional aircraft treat self-weight and payload as a continuous burden; within this system, total vehicle mass becomes the energy source of the descent phase. Condensed statement:

> 重力越大，发电能力越强。 The greater the gravity, the greater the power generation capacity.

3.2.4 多旋翼构型在能量回收中的先天优势

3.2.4 Innate Advantages of the Multi-Rotor Configuration in Energy Recovery

多旋翼分布式驱动架构相比单旋翼平台，在动能回收层面具备不可替代优势： 常规直升机局限： 下降依靠桨叶变距控速，传动系统原生面向动力输出设计；即便改造实现风车发电，整体能量转换效率偏低。

Compared with single-rotor platforms, the multi-rotor distributed drive architecture has irreplaceable advantages in kinetic-energy recovery: Limitation of conventional helicopters: Descent speed is controlled by blade pitch, and the transmission is natively designed for power output; even if modified for windmilling generation, the overall energy-conversion efficiency is low.

多旋翼平台优势：

Advantages of the multi-rotor platform:

电机直驱桨叶，动力链路简洁，支持正向驱动、反向发电双向工况切换；

Motors drive the blades directly, with a simple power train supporting two-way switching between forward drive and reverse generation;

下降气流驱动桨叶自转时，电机可无缝切换至整流发电模式；

When the descending airflow drives blade rotation, the motors switch seamlessly into rectified generating mode;

分布式布局实现全域同步能量捕获，总回收功率远高于单点回收方案。

The distributed layout achieves synchronous energy capture over the whole platform, with total recovered power far exceeding single-point recovery schemes.

多旋翼直驱方案天然具备能量回收硬件基础，无需大规模机械改造。

The multi-rotor direct-drive scheme inherently provides the hardware basis for energy recovery without large mechanical modifications.

3.3 再入阶段叶片滑翔运行策略

3.3 Blade-Gliding Operating Strategy During Re-entry

当平台自亚轨道、近地轨道返回大气层时，将面临高速再入气动热难题。传统运载航天器依靠厚重烧蚀隔热盾抵御高温。本系统采用差异化控制方案。

When the platform returns from suborbital or low-Earth orbit into the atmosphere, it faces the problem of high-speed re-entry aerodynamic heating. Conventional launch spacecraft rely on heavy ablative heat shields to withstand the high temperature. This system adopts a differentiated control scheme.

3.3.1 侧向滑翔弹道设计

3.3.1 Lateral Gliding Trajectory Design

平台再入弹道不采用近乎垂直的陡直坠落路线，采取侧向平缓滑翔，运动形态类比叶片缓慢飘落，沿倾斜轨迹渐进渗透大气层。

The platform's re-entry trajectory does not follow a near-vertical steep drop but takes a gentle lateral glide, moving like a leaf slowly drifting down, penetrating the atmosphere progressively along an inclined path.

滑翔弹道特征：

Features of the gliding trajectory:

再入初始姿态相对水平面维持 10°~30° 小倾角；

The initial re-entry attitude maintains a small inclination of 10°–30° relative to the horizontal;

阻力面板完全展开，构建大面积气动减速面；

The drag panels are fully deployed, forming a large-area aerodynamic deceleration surface;

大气阻力持续消耗轨道动能，将 7.8km/s 轨道速度逐步降低至亚音速区间；

Atmospheric drag continuously consumes orbital kinetic energy, reducing the 7.8 km/s orbital speed step by step into the subsonic range;

全程减速时间更长，依靠更长飞行路径平缓消耗动能。

The deceleration lasts longer overall, consuming kinetic energy gently over a longer flight path.

3.3.2 降低热负荷原理

3.3.2 Principle of Reducing Thermal Load

传统航天器再入高温诱因：弹道陡峭、减速集中在短时间、大气密度快速上升，产生极高峰值热流密度。

Causes of the high temperature in conventional spacecraft re-entry: steep trajectory, deceleration concentrated in a short time, and rapid rise of air density, producing an extremely high peak heat flux.

叶片滑翔方案拉长减速区间，使平台渐进穿越不同高度大气层，大气密度缓慢提升，气动加热峰值显著下降。 同时，完全展开的阻力面板拥有巨大散热表面积，摩擦热量均匀分布；面板间隙形成对流通道，避免局部热量持续聚集。

The blade-glide scheme stretches the deceleration interval, letting the platform pass through the atmosphere at different altitudes progressively; air density rises slowly and the aerodynamic-heating peak drops markedly. At the same time, the fully deployed drag panels offer a huge heat-dissipating surface, with frictional heat distributed evenly; the gaps between panels form convection channels, preventing localized continuous heat accumulation.

系统再入设计思路总结：不依靠结构硬抗极端热载荷，依靠时间换取热流强度降低。

Summary of the system's re-entry design philosophy: do not resist extreme thermal loads with structure, but trade time for reduced heat flux.

> 补充说明：该方案在极高轨道速度工况下仍存在气动热挑战，并非完全消除加热效应，仅显著降低峰值热负荷，削弱对重型隔热结构的依赖。 Note: at extremely high orbital speeds aerothermal challenges still remain; this scheme does not eliminate heating entirely, but markedly lowers the peak thermal load and weakens reliance on heavy thermal-protection structures.

3.3.3 磁阻力辅助卸荷机制

3.3.3 Magnetic-Drag Auxiliary Load-Shedding Mechanism

叶片滑翔全过程，电磁阻尼系统持续参与工作，依托电磁感应消耗动能发电。附加增益效果：电磁阻尼提供非接触式制动力，分担气动阻力施加在框架上的机械载荷。

Throughout the blade-glide process the electromagnetic damping system keeps working, consuming kinetic energy to generate power via electromagnetic induction. Additional benefit: electromagnetic damping provides non-contact

braking force, sharing the mechanical load that aerodynamic drag applies to the frame.

作用效果：部分减速载荷由电磁场承担，降低飞行器框架峰值应力，有利于实现轻量化结构设计。

Effect: part of the deceleration load is borne by the electromagnetic field, lowering the peak stress on the airframe, which favors lightweight structural design.

3.3.4 再入模式方案对比

3.3.4 Comparison of Re-Entry Modes

对比项 Comparison item	传统航天器弹道式再入 Traditional spacecraft ballistic re-entry	本文叶片滑翔再入方案 Blade-glide re-entry scheme of this paper
下降弹道 Descent trajectory	陡峭、接近垂直 Steep, near-vertical	平缓侧向滑翔轨迹 Gentle lateral gliding trajectory
速度变化特征 Speed-change characteristic	短时急剧减速 Sharp deceleration in short time	长时渐进减速 Long, progressive deceleration
气动热负荷 Aerothermal load	峰值极高，必须配置厚重隔热盾 Extremely high peak, heavy heat shield mandatory	峰值显著降低，可简化热防护结构 Peak markedly lowered, thermal protection simplified
过程能量回收 Energy recovery during the process	无能量回收手段 No energy-recovery means	全程持续捕获动能发电 Continuous kinetic-energy capture and generation throughout
结构承受应力 Stress borne by structure	瞬时峰值载荷大 Large instantaneous peak load	载荷平缓，电磁阻尼分担应力 Smooth loading, electromagnetic damping shares stress
飞行可控性 Flight controllability	黑障区间通信受限，容错窗口小 Communication limited in the blackout zone, small fault-tolerance window	减速过程平缓，姿态调控窗口更大 Gentle deceleration, larger attitude-control window

第 4 章 多旋翼网状拼接结构设计

Chapter 4 Multi-Rotor Mesh-Splicing Structural Design

本章详细阐述核心构型：网状模块化运载平台的整体架构、冗余容错机理与能源系统布局约束。

This chapter details the core configuration: the overall architecture of the mesh modular transport platform, the redundancy fault-tolerance mechanism, and the constraints on energy-system layout.

4.1 网状拼接架构方案

4.1 Mesh-Splicing Architecture

平台动力系统由大量标准化动力单元（涵道旋翼 / 倾转旋翼）通过桁架框架相互拼接组网。每一套动力单元均为具备完整功能的独立飞行模块：

The platform's power system consists of many standardized power units (ducted rotors / tiltrotors) interconnected into a mesh through a truss frame. Each power unit is an independent flight module with complete functionality:

独立动力组件：电机、桨叶、电子调速器；

Independent power assembly: motor, blades, electronic speed controller;

独立感知控制：机载芯片、惯性、气压传感模块；

Independent sensing and control: onboard chip, inertial and barometric sensing modules;

独立能源接口：储能总线、电源管理单元；

Independent energy interface: energy-storage bus and power-management unit;

独立通信单元：模块间组网通信、与主控交互链路。

Independent communication unit: inter-module networking and the link to the central controller.

大量单元完成物理组网后，由中央主控单元统一协调推力、姿态，整合形成大尺寸一体化升降运载平台。

After the units are physically networked, a central controller coordinates thrust and attitude uniformly, integrating them into one large-scale lift-transport platform.

4.2 冗余容错运行机理

4.2 Redundancy Fault-Tolerance Operating Mechanism

网状模块化构型最核心优势为高故障冗余能力。 常规飞行器属于一体化串联体系，关键部件失效极易引发整机失控；网状平台采用分布式并联架构，局部单元故障不直接威胁整体稳定。

The most core advantage of the mesh modular configuration is its high fault redundancy. Conventional aircraft are integrated serial systems in which failure of a key component easily causes loss of control of the whole vehicle; the mesh platform uses a distributed parallel architecture in which local unit failures do not directly threaten overall stability.

运行逻辑示例：

Example of the operating logic:

平台由 18 组动力单元组网构成；

The platform is networked from 18 power units;

飞行途中 2~3 组单元遭遇撞击、器件故障完全失效；

During flight, 2–3 units fail completely due to impact or device faults;

主控实时识别失效单元，将故障模块从推力分配体系剔除；

The central controller identifies the failed units in real time and removes the faulty modules from the thrust-allocation system;

剩余正常单元自动提升输出功率、微调姿态角，补偿损失升力；

The remaining healthy units automatically raise output power and fine-tune attitude angles to compensate for the lost lift;

平台整体姿态维持稳定，可持续完成剩余飞行任务。

The platform's overall attitude remains stable and the remaining flight task can still be completed.

容错能力量化说明： 设单一动力单元单次飞行失效概率为，总单元数量，系统失效临界阈值为同时故障单元数量。 多单元同时失效概率满足二项分布：

Quantitative description of fault tolerance: Let the failure probability of a single power unit per flight be, the total number of units, and the system's failure threshold be the number of simultaneously failed units. The probability of multi-unit simultaneous failure follows the binomial distribution:

当组网单元数量充足（如），即便单一模块故障率达到 5%，多模块同步失效引发整机失控的概率仍维持在较低水平。 通俗表述：少量单元损坏不影响飞行；只有故障单元数量突破冗余阈值，平台才会丧失稳定运行能力。

When the number of networked units is sufficient (e.g.,), even if a single module's failure rate reaches 5%, the probability of whole-vehicle loss of control caused by simultaneous multi-module failure remains low. In plain terms: a few damaged units do not affect flight; only when the number of failed units exceeds the redundancy threshold does the platform lose its ability to operate stably.

4.3 积木式模块化运维优势

4.3 Advantages of Building-Block Modular Operation and Maintenance

网状拼接构型具备传统整机飞行器不具备的运维特性：模块化快速更换。 传统飞行器结构受损通常需要整机返厂大修，严重损伤条件下直接整机报废。本平台运维模式：

The mesh-splicing configuration has an O&M property that conventional integrated aircraft lack: rapid modular replacement. When a conventional aircraft's structure is damaged it usually needs a factory-level overhaul, and severe damage can lead to scrapping the whole vehicle. The O&M model of this platform:

故障单元可直接自桁架框架上拆卸；

Failed units can be detached directly from the truss frame;

替换全新标准化单元，完成机械与电气连接即可恢复功能；

A new standardized unit is installed, and once the mechanical and electrical connections are made, full function is restored;

无需整机拆解大修，条件允许时甚至可实现空中在轨模块更换。

No whole-vehicle disassembly is needed; if conditions allow, even in-flight on-orbit module replacement is possible.

积木式模块化带来三层价值：

Building-block modularity brings three layers of value:

维护成本降低：仅更换受损单元，完好模块持续复用；

Lower maintenance cost: only damaged units are replaced while intact modules keep being reused;

整机服役周期延长：主体桁架结构完好前提下，动力单元可持续迭代更换；

Longer service life: as long as the main truss structure is sound, power units can be continuously upgraded and replaced;

任务灵活拓展：依据载荷指标随时增加、减少动力单元数量。

Flexible mission scaling: the number of power units can be increased or decreased at any time according to payload requirements.

4.4 能源系统布局与运行约束

4.4 Energy-System Layout and Operating Constraints

4.4.1 电能作为首选动力形式

4.4.1 Electrical Energy as the Preferred Power Form

本系统优先采用电能作为动力源，电能体系优势：

This system prefers electrical energy as its power source; the advantages of an electric system:

稳态工作热损耗可控，支持长时间持续飞行；

Controllable steady-state thermal loss, supporting long-duration continuous flight;

运行噪声水平低，适配多样化空域环境；

Low operating noise, fitting diverse airspace environments;

推力调控响应速度快，满足多单元协同精细姿态控制；

Fast thrust-regulation response, meeting fine multi-unit coordinated attitude control;

管路系统简单，相比化学推进系统维护难度更低。

Simple piping systems, easier maintenance than chemical propulsion systems.

固有短板：储能电池自重较大，额外增加爬升阶段运载总质量，一定程度抵消减阻收益。该矛盾属于典型工程权衡问题。

Inherent shortcoming: storage batteries are heavy, adding to the total transported mass during climb and partly offsetting the drag-reduction gains. This contradiction is a typical engineering trade-off.

4.4.2 电池自重的工程迭代思路

4.4.2 Iterative Engineering Approach to Battery Weight

电池自重带来的利弊权衡无法依靠理论推演获得唯一最优解，只能通过迭代试验优化：

The trade-off brought by battery weight cannot be resolved into a unique optimum by theoretical deduction alone; it can only be optimized through iterative testing:

测试不同储能容量配置下平台爬升能耗；

Measure climb energy consumption under different energy-storage configurations;

测试同等配置下降阶段电能回收总量；

Measure the total electrical energy recovered during descent under the same configuration;

绘制「电池自重 — 系统净能耗」变化曲线；

Plot the “battery weight — system net energy consumption” curve;

选取净能耗最低的储能配置作为该任务下最优方案。

Select the energy-storage configuration with the lowest net energy consumption as the optimum for that mission.

最优储能参数随电池能量密度技术迭代持续变化，本文不给出固定最优电池容量数值，该参数属于动态工程优化目标，并非固定物理常数。

The optimal storage parameters keep changing as battery energy-density technology evolves; this paper gives no fixed optimal battery capacity value — this parameter is a dynamic engineering optimization target, not a fixed physical constant.

4.4.3 能量闭环运行硬性约束

4.4.3 Hard Constraint of the Closed-Loop Energy Operation

平台实现多次自主起降、能量自循环的基础约束条件： 爬升阶段消耗总电能，不得超过系统总储能容量的 2/3。

The basic constraint for the platform to achieve repeated autonomous takeoffs/landings and a self-sustaining energy cycle: The total electrical energy consumed during climb must not exceed 2/3 of the system's total storage capacity.

预留 1/3 储能作为安全冗余，覆盖下述工况需求：

The reserved 1/3 of storage serves as safety redundancy, covering the following needs:

突发大气紊流扰动下姿态稳定、航向修正能耗；

Energy for attitude stabilization and course correction under sudden atmospheric turbulence;

局部动力单元故障后的升力补偿能耗；

Lift-compensation energy after local power-unit failures;

环境变化导致下降路径延长带来的额外能耗；

Extra energy when environmental changes lengthen the descent path;

连续多次起降循环的能量缓冲。

An energy buffer for consecutive takeoff-and-landing cycles.

若爬升能耗超过储能总量 2/3，即便下降阶段能量回收效率达到理论上限，也无法将储能恢复至安全起飞标准。

If climb consumption exceeds 2/3 of total storage, even a theoretically maximal descent energy recovery cannot recharge the system to the safe takeoff standard.

2/3 储能占比属于工程安全红线。 工程设计目标应尽可能降低爬升能耗占比，扩大能量冗余裕度。

The 2/3 storage ratio is an engineering safety red line. Design goals should minimize the climb energy ratio as far as possible, widening the energy-redundancy margin.

第 5 章 系统性能理论分析

Chapter 5 Theoretical Performance Analysis of the System

5.1 基础空气动力学公式

5.1 Basic Aerodynamic Formulas

全文气动性能推演依托经典空气阻力方程： 参数定义： — 气动阻力； — 空气密度； — 相对气流速度； — 阻力系数； — 迎风参考面积。

The aerodynamic performance deduction throughout the paper rests on the classical drag equation: Parameter definitions: — aerodynamic drag; — air density; — relative airflow speed; — drag coefficient; — reference frontal area.

5.2 上升阶段性能推演

5.2 Performance Deduction of the Ascent Phase

爬升工况阻力面板收拢，阻力系数显著降低。 参数对比参考： 常规刚性运载平台 ； 本系统收拢构型 ； 气动阻力降幅:

During climb the drag panels are retracted and the drag coefficient drops markedly. Reference parameter comparison: Conventional rigid transport platform ; This system's retracted configuration ; Reduction of aerodynamic drag:

爬升全程能耗表达式： 第一项为重力势能增量，第二项为克服全程气动阻力消耗能量。

Energy-consumption expression for the whole climb: The first term is the increment of gravitational potential energy; the second term is the energy spent overcoming aerodynamic drag over the whole path.

5.3 下降阶段性能推演

5.3 Performance Deduction of the Descent Phase

下降工况阻力面板完全展开，最高可趋近 1.0。 瞬时发电功率:

During descent the drag panels are fully deployed, and the reaches at most close to 1.0. Instantaneous generating power:

能量回收效率定义:

Definition of energy-recovery efficiency:

缩比试验数据显示，理想全开工况下动能回收效率可达 80%~90%。

Scaled-model test data show that under the ideal fully open condition the kinetic-energy recovery efficiency can reach 80%–90%.

5.4 惯性冲出稠密大气层能量分析

5.4 Energy Analysis of Inertial Run-Out Through the Dense Atmosphere

平台一项重要能力：依靠爬升积累动能，惯性滑行脱离低层稠密大气。

An important capability of the platform: relying on the kinetic energy accumulated during climb, it coasts inertially out of the low dense atmosphere.

工作机理： 低空区间平台持续消耗能量克服重力与气动阻力；随高度提升，大气密度随高度指数衰减： 大气标高取值约 8.5km。

Working mechanism: In the low-altitude region the platform continuously spends energy overcoming gravity and aerodynamic drag; as altitude increases, atmospheric density decays exponentially with height: The atmospheric scale height is about 8.5 km.

高度达到 20~30km 后，空气密度大幅衰减，气动阻力影响基本可以忽略。平台无需持续输出推力对抗阻力，依靠已有动能持续向上惯性滑行。

Above 20–30 km the air density has decayed drastically and aerodynamic drag is essentially negligible. The platform no longer needs continuous thrust to fight drag, coasting upward on its existing kinetic energy.

在平流层顶端（约 50km 高度），大气密度仅为地面水平的 0.3% 左右。平台依靠惯性继续滑行直至临近卡门线（100km）。

At the top of the stratosphere (about 50 km) air density is only about 0.3% of the sea-level value. The platform continues coasting on inertia until near the Kármán line (100 km).

该过程本质：爬升阶段积蓄动能持续转化为重力势能，后期不再需要持续动力输入。

The essence of this process: the kinetic energy stored during climb is continuously converted into gravitational potential energy, with no sustained power input needed in the later stage.

5.5 能量闭环判定准则

5.5 Criterion for the Closed-Loop Energy Cycle

综合全程能量平衡关系，定义净能量指标： 式中： — 爬升阶段总耗电量； — 下降阶段回收电能总量。

Combining the energy balance over the whole flight, the net-energy index is defined as: where: — total electrical energy consumed during climb; — total electrical energy recovered during descent.

判定标准：当，平台具备理论自维持循环运行能力。

Criterion: when , the platform possesses a theoretical self-sustaining cyclic operating capability.

试验推演表明：常规中等载荷、中等高度工况下存在满足的工作区间；载荷过大、爬升高度超过临界值时，将转变为能量净消耗工况。

Test deduction shows that for normal medium-payload, medium-altitude conditions a working region satisfying exists; when the payload is too large or the climb altitude exceeds the critical value, the case turns into net energy consumption.

第 6 章 试验验证方案

Chapter 6 Experimental Verification Plan

6.1 两级试验体系设计

6.1 Design of the Two-Tier Test System

整套验证方案划分为单元部件试验与整机系统联试两个层级。 单元部件试验：

The whole verification plan is divided into two tiers: component-level tests and full-system integration tests. Component-level tests:

倾转、涵道两类动力单元单独动力特性标定；

Individual power-characteristic calibration of the tiltrotor and ducted-rotor units;

阻力面板展开 / 收拢机构响应速度、结构强度测试；

Response-speed and structural-strength tests of the drag-panel deploy/retract mechanism;

不同模拟风速下电机风车工况发电功率标定。

Calibration of motor windmilling generating power under different simulated wind speeds.

系统集成试验：

System-integration tests:

搭建 4~6 单元小型网状缩比验证平台；

Build a small 4–6-unit scaled mesh verification platform;

实测爬升阶段能耗、下降阶段电能回收总量；

Measure climb-phase energy consumption and descent-phase total energy recovery;

人为关闭部分动力单元，测试平台故障容错性能。

Deliberately shut down some power units to test the platform's fault tolerance.

6.2 关键预期试验结果汇总

6.2 Summary of Key Expected Test Results

测试项目 Test item	预期指标 Expected target	说明 Remarks
爬升气动阻力降幅 Climb aerodynamic-drag reduction	40%~60% 40%~60%	网状构型 + 阻力面板收拢状态 Mesh configuration + retracted drag panels
下降动能回收效率 Kinetic-energy recovery efficiency during descent	80%~90% 80%~90%	阻力面板完全展开工况 Drag panels fully deployed
常规工况净能量状态 Net-energy state under normal conditions	正值（发电量＞爬升耗电） Positive (generation > climb consumption)	中等载荷、中等爬升高度区间 Medium payload, medium climb-altitude band
极限工况净能量状态 Net-energy state under extreme conditions	负值（发电量＜爬升耗电） Negative (generation < climb consumption)	超载、极限高度任务场景 Overload and extreme-altitude mission scenarios
故障容错阈值 Fault-tolerance threshold	容忍 10%~20% 动力单元失效 Tolerates 10%~20% power-unit failure	剩余正常单元补偿升力维持稳定 Remaining healthy units compensate lift and keep stability

6.3 工程迭代思路

6.3 Engineering Iteration Approach

需要明确：平台各项最优运行参数属于工程迭代优化问题，不属于基础架构理论问题。 储能容量、电机功率、面板面积、组网单元数量的最优组合高度依赖任务场景。推荐迭代路径：

It must be made clear that the platform's optimal operating parameters are an engineering-iteration optimization problem, not a basic-architecture theory problem. The optimal combination of storage capacity, motor power, panel area and number of networked units depends heavily on the mission scenario. The recommended iteration path:

制造多组不同参数配置缩比样机；

Build several groups of scaled prototypes with different parameter configurations;

分组开展升降循环试验，采集能耗、发电数据；

Run grouped ascent-descent cycle tests, collecting energy-consumption and power-generation data;

横向对比试验结果，筛选匹配目标任务的最优参数组合。

Compare test results horizontally and select the parameter combination best matching the target mission.

参数优化迭代不影响核心架构可行性；基础方案框架确立后，参数调优属于渐进式工程完善工作。

Parameter-optimization iterations do not affect the feasibility of the core architecture; once the basic framework is established, parameter tuning is progressive engineering refinement.

上篇小结

Part I Summary

本文提出一套依托空气阻力双向收割机制的多旋翼网状拼接升降运载飞行器系统，主要结论汇总如下：

This paper presents a multi-rotor mesh-spliced lift-transport aircraft system based on bidirectional air-drag harvesting. The main conclusions are summarized below:

空气阻力具备双向资源化利用潜力。 爬升阶段收拢机构抑制阻力降低能耗，下降阶段展开面板捕获气流动能发电，实现空气阻力由单向损耗因素向双向可利用资源转变。

Air drag has bidirectional resource-utilization potential. During climb the retraction mechanism suppresses drag and cuts energy consumption; during descent the deployed panels capture airflow kinetic energy for generation, turning air drag from a one-way loss factor into a bidirectional usable resource.

网状模块化构型提供高冗余容错能力。 动力单元并联组网，局部模块失效不会直接造成整机失控；积木式可拆卸结构大幅降低运维成本，延长平台服役周期。

The mesh modular configuration provides high redundancy fault tolerance. Power units are networked in parallel; local module failures do not directly cause loss of control of the whole vehicle; the building-block detachable structure greatly lowers O&M cost and extends the platform's service life.

系统存在独特载荷特性：总重力越大，下降发电能力越强。 该特性使平台在重载运载场景下具备相较传统飞行器的能量优势。

The system has a unique payload property: the greater the total gravity, the stronger the descent power generation. This property gives the platform an energy advantage over conventional aircraft in heavy-payload transport.

叶片侧向滑翔再入方案可降低再入热负荷。 通过延长滑翔路径、平缓减速，分散气动加热峰值，简化航天器所需热防护结构。

The blade-like lateral gliding re-entry scheme can lower re-entry thermal load. By lengthening the glide path and decelerating gently, the aerodynamic-heating peak is spread out, simplifying the thermal-protection structure required of the spacecraft.

能量自闭环运行具备理论可行性。 在合适载荷与高度区间内，下降回收电能可覆盖爬升阶段大部分能耗，大幅降低外部持续供能需求。

Self-closed energy-loop operation is theoretically feasible. Within suitable payload and altitude ranges, the electricity recovered during descent can cover most of the climb consumption, greatly reducing the need for continuous external energy supply.

本方案的核心价值在于提供一套非化学喷射推进、具备能量循环潜力的可重复升降运载范式。该系统并非现有飞行器的局部改良方案，提供了高空运载、航天前置助推领域全新的技术路线。

The core value of this scheme lies in providing a reusable lift-transport paradigm with no chemical jet propulsion and with energy-cycle potential. The system is not a local improvement of existing aircraft; it offers a new technical route for high-altitude transport and space pre-boosting.

（上篇完）

(End of Part I)

下篇：结构设计、能量管理、应用场景与系统对比

Part II: Structural Design, Energy Management, Application Scenarios and System Comparison

第 7 章 结构设计：阻力面板与网状拼接

Chapter 7 Structural Design: Drag Panels and Mesh Splicing

7.1 阻力面板的结构形式

7.1 Structural Forms of the Drag Panels

阻力面板是下降阶段能量回收的核心执行部件。本系统采用两种可选的阻力面板结构形式：花瓣式展开结构和百叶窗式翻转结构。两种形式各有特点，可根据任务场景选择使用。

The drag panels are the core actuators of energy recovery during descent. This system adopts two optional drag-panel structural forms: the petal-type deployment structure and the louver-type flipping structure. Each has its own characteristics and can be selected according to the mission scenario.

7.1.1 花瓣式展开结构

7.1.1 Petal-Type Deployment Structure

花瓣式展开阻力面板的设计灵感来源于自然界中花朵的开放机制。多个扇形面板通过铰链连接至飞行器主体，在下降阶段向外展开，形成类似伞状或花朵状的大面积阻力面。

The design of the petal-type deployable drag panel is inspired by the blooming mechanism of flowers in nature. Multiple fan-shaped panels are hinged to the aircraft body and swing outward during descent, forming a large-area drag surface resembling an umbrella or a blossom.

结构特点：

Structural features:

由 8-16 个扇形面板组成，围绕飞行器主体呈环形排列

Composed of 8–16 fan-shaped panels arranged ring-wise around the aircraft body

每个面板独立驱动，可单独控制展开角度

Each panel is independently driven, and its deployment angle can be controlled separately

面板采用轻质复合材料（碳纤维或芳纶蜂窝结构）

The panels use lightweight composite materials (carbon fiber or aramid honeycomb structure)

展开后整体呈圆形或梅花形，迎风面积可增加 3-5 倍

After deployment the overall shape is circular or plum-blossom-like, increasing the frontal area by 3–5 times

收起时面板折叠贴合机身，上升阶段保持流线型外形

When stowed, the panels fold flush against the fuselage, keeping a streamlined shape during ascent

优势：

Advantages:

展开后阻力面积大，空气阻力捕获效率高

Large drag area after deployment, high air-drag capture efficiency

受力均匀，结构应力分布合理

Even force distribution with a reasonable structural stress layout

展开过程具有自稳定性（气动中心与质心自动对齐）

Self-stabilizing deployment (the aerodynamic center aligns automatically with the center of mass)

视觉识别度高，便于地面观测和遥控

High visual recognizability, convenient for ground observation and remote control

单面板损坏不影响其他面板的展开与工作

Damage to a single panel does not affect the deployment and operation of the others

劣势：

Disadvantages:

收起状态下仍占用一定空间，不能完全贴合机身

Still occupies some space when stowed and cannot lie fully flush against the fuselage

对侧向风较为敏感，强风环境下可能需要收窄

Fairly sensitive to crosswinds; in strong-wind environments it may need to narrow its deployment

制造工艺相对复杂，铰链数量多

Relatively complex manufacturing process with many hinges

需要精确的同步控制，避免展开不均匀

Requires precise synchronous control to avoid uneven deployment

适用场景： 需要较大发电功率的任务，或下降路径垂直度较高的场景。

Applicable scenarios: missions requiring larger generating power, or descents with high trajectory verticality.

7.1.2 百叶窗式翻转结构

7.1.2 Louver-Type Flipping Structure

百叶窗式翻转阻力面板由多个条形叶片平行排列，通过翻转机构实现 "打开"（产生阻力）和 "关闭"（减少阻力）的状态切换。

The louver-type flipping drag panel consists of multiple strip-like blades arranged in parallel, switching between the “open” (generating drag) and “closed” (reducing drag) states through a flipping mechanism.

结构特点：

Structural features:

由 10-20 个条形叶片组成，平行排列

Composed of 10–20 strip-like blades arranged in parallel

叶片通过连杆机构联动翻转

The blades flip in linkage through a rod mechanism

叶片关闭时贴合机身表面，形成流线型

When closed, the blades lie flush against the fuselage surface, forming a streamlined shape

叶片打开时垂直于气流方向，产生最大阻力

When open, the blades stand perpendicular to the airflow, producing maximum drag

每个叶片可独立调节翻转角度，实现阻力的精细化控制

Each blade can independently adjust its flip angle, achieving fine-grained drag control

优势：

Advantages:

收起状态贴合机身，上升阶段阻力极小

Flush against the fuselage when stowed, with minimal drag during ascent

结构简单，可靠性高

Simple structure with high reliability

叶片可逐级打开，实现阻力的渐进式调节

Blades can open stage by stage, achieving progressive drag regulation

制造和维护成本较低

Lower manufacturing and maintenance cost

叶片之间的间隙允许气流通过，降低湍流噪声

Gaps between blades let airflow pass, reducing turbulence noise

劣势：

Disadvantages:

展开后阻力面积有限，总阻力小于花瓣式

Limited deployed drag area; total drag smaller than the petal type

叶片之间存在气流泄漏，能量捕获效率略低

Airflow leakage between blades slightly lowers energy-capture efficiency

叶片旋转轴需要高耐久性设计（频繁翻转）

Blade pivot axes need high-durability design (frequent flipping)

在高速气流中，叶片可能产生振动（需阻尼设计）

In high-speed airflow the blades may vibrate (damping design required)

适用场景： 需要频繁起降、或对机身流线型要求较高的任务。

Applicable scenarios: missions with frequent takeoffs and landings, or with high streamlined-fuselage requirements.

7.1.3 两种结构的对比与选择建议

7.1.3 Comparison of the Two Structures and Selection Advice

特性 Property	花瓣式 Petal type	百叶窗式 Louver type
最大阻力面积 Maximum drag area	大（3-5 倍迎风面积） Large (3-5x frontal area)	中（2-3 倍迎风面积） Medium (2-3x frontal area)
	85%-90% 85%-90%	75%-85% 75%-85%

能量回收效率 Energy-recovery efficiency		
收起状态流线型 Streamlining when stowed	良 Good	优 Excellent
结构复杂度 Structural complexity	较高 Higher	较低 Lower
制造成本 Manufacturing cost	较高 Higher	较低 Lower
维护难度 Maintenance difficulty	中等 Medium	低 Low
强风适应性 Strong-wind adaptability	较敏感 Fairly sensitive	较稳定 Fairly stable
故障容错 Fault tolerance	单面板损坏不影响整体 Damage to a single panel does not affect the whole	单叶片损坏可通过锁止隔离 A damaged blade can be locked and isolated
推荐场景 Recommended scenario	深空返回、重载下降 Deep-space return, heavy-load descent	常规起降、城际飞行 Regular takeoff/landing, inter-city flight

推荐方案： 对于多数任务场景，建议优先选择百叶窗式翻转结构，因其结构简单、可靠性高、维护成本低。对于需要最大能量回收的深空或重载任务，可选择花瓣式展开结构。

Recommended scheme: for most mission scenarios, the louver-type flipping structure is preferred for its simple structure, high reliability and low maintenance cost; for deep-space or heavy-payload missions requiring maximum energy recovery, the petal-type deployment structure can be chosen.

7.1.4 阻力面板的材料选择

7.1.4 Material Selection for the Drag Panels

阻力面板需要在轻量化、强度、耐热性和成本之间取得平衡。

The drag panels must balance light weight, strength, heat resistance and cost.

材料 Material	密度(g/cm³) Density (g/cm³)	比强度 Specific strength	耐温(°C) Heat resistance (°C)	成本 Cost	适用部位 Applicable location

碳纤维复合材料 Carbon-fiber composite	1.6 1.6	极高 Extremely high	150-200 150-200	高 High	主结构、铰链 Primary structure, hinges
芳纶蜂窝夹层 Aramid honeycomb sandwich	1.4 1.4	高 High	100-150 100-150	中 Medium	面板主体 Panel body
玻璃纤维复合材料 Glass-fiber composite	1.8 1.8	中 Medium	100-150 100-150	低 Low	次承力件 Secondary load-bearing parts
钛合金 Titanium alloy	4.5 4.5	中 Medium	500+ 500+	很高 Very high	高温区铰链 Hinges in high-temperature zones

推荐方案： 面板主体采用芳纶蜂窝夹层结构（轻质、高韧性）；铰链和连接件采用钛合金（耐高温、抗疲劳）；碳纤维作为主框架材料（高强度、低重量）。

Recommended scheme: the panel body uses aramid honeycomb sandwich structure (light, high toughness); hinges and connectors use titanium alloy (high-temperature resistant, fatigue resistant); carbon fiber serves as the main frame material (high strength, low weight).

7.2 网状拼接结构的连接方式

7.2 Connection Methods of the Mesh-Splicing Structure

本系统的网状拼接结构需要解决一个核心工程问题：多个独立的旋翼单元如何牢固地连接在一起，同时保持各自的工作独立性?

The mesh-splicing structure of this system must solve one core engineering problem: how can multiple independent rotor units be firmly connected together while each keeps its operational independence?

7.2.1 机械连接方式

7.2.1 Mechanical Connection

采用框架式连接方案:

A frame-type connection scheme is adopted:

每个旋翼单元通过 4-6 个连接点与主框架连接

Each rotor unit connects to the main frame through 4–6 connection points

连接点采用快拆锁紧机构，支持快速安装和拆卸

The connection points use quick-release locking mechanisms, supporting rapid installation and removal

主框架为轻质铝合金或碳纤维复合材料桁架结构

The main frame is a lightweight aluminum-alloy or carbon-fiber-composite truss structure

单元与单元之间预留间隙，避免气流相互干扰

Gaps are reserved between units to avoid mutual airflow interference

整体结构呈蜂窝状或三角形网格布局，确保力的均匀传递

The overall structure forms a honeycomb-like or triangular mesh layout, ensuring even force transmission

框架设计的核心原则：

Core principles of frame design:

足够牢固：承受飞行中的气动载荷和重力载荷

Strong enough: to withstand aerodynamic and gravitational loads in flight

尽量轻量：不成为系统重量的主要来源

As light as possible: not becoming the main source of system weight

便于拆装：支持快速更换单元

Easy to assemble and disassemble: supporting rapid unit replacement

抗疲劳：承受多次起降的循环载荷

Fatigue-resistant: to endure the cyclic loads of repeated takeoffs and landings

7.2.2 电气与控制连接

7.2.2 Electrical and Control Connections

每个单元通过标准化接口（电源线 + 通信线）接入主系统

Each unit joins the main system through a standardized interface (power lines + communication lines)

通信协议采用冗余双总线架构（主总线和备用总线）

The communication protocol adopts a redundant dual-bus architecture (main bus and backup bus)

支持热插拔：故障单元可在系统运行状态下更换

Hot swapping is supported: failed units can be replaced while the system is running

每个单元独立供电和独立控制，不依赖其他单元

Each unit is independently powered and controlled, not relying on other units

电源线采用环形冗余布线：即使某段线路断裂，仍可通过其他路径供电

Power lines use ring redundant wiring: even if one segment breaks, supply continues through other paths

7.2.3 连接强度与重量平衡

7.2.3 Balancing Connection Strength and Weight

连接点的设计需在 "足够牢固" 和 "尽量轻量" 之间寻找平衡

The design of connection points must find a balance between “strong enough” and “as light as possible”

有限元分析（FEA）用于优化连接点的形状和材料厚度

Finite element analysis (FEA) is used to optimize the shape and material thickness of the connection points

轻量化目标：连接结构总重量不超过系统总重量的 10%-15%

Lightweight target: the total weight of the connection structure shall not exceed 10%–15% of the system's total weight

关键连接点采用双保险设计（主连接 + 备用连接）

Key connection points adopt a double-insurance design (primary connection + backup connection)

7.2.4 网状结构的维修性与抗损性优势

7.2.4 Maintainability and Damage-Resistance Advantages of the Mesh Structure

多旋翼网状拼接结构在维护和生存能力上，与传统一体式飞行器有本质区别：

The multi-rotor mesh-splicing structure differs fundamentally from conventional integrated aircraft in maintenance and survivability:

1. 独立更换，无需整机维修

1. Independent replacement without whole-vehicle repair

传统飞行器结构一旦损坏，往往需要大面积拆解、返厂维修，甚至整机报废。本系统的网状拼接结构支持独立模块更换 —— 任何一个旋翼单元失效或受损，只需单独将该单元从网状框架上拆下，换入新的单元即可恢复完整功能。整个维修过程不涉及其他单元，不需要整机拆卸，显著降低了维护难度和时间成本。

When a conventional aircraft's structure is damaged, it often requires extensive disassembly, factory repair, or even scrapping of the whole vehicle. The mesh-splicing structure of this system supports independent module replacement — whenever any rotor unit fails or is damaged, that unit alone is detached from the mesh frame and a new unit is installed to restore full function. The whole repair involves no other unit and needs no vehicle-level disassembly, greatly reducing maintenance difficulty and time cost.

2. 部分单元失效，不影响整体功能

2. Partial unit failure does not affect overall function

网状结构的核心优势在于其高冗余度。即使有部分单元（如旋翼、阻力面板模块）在飞行中失效或脱落，其余单元仍可通过动态调整推力和姿态，维持平台的稳定飞行。系统不会因为局部损伤而整体失效。

The core advantage of the mesh structure lies in its high redundancy. Even if some units (such as rotors or drag-panel modules) fail or detach during flight, the remaining units can maintain stable flight of the platform by dynamically adjusting thrust and attitude. The system does not fail as a whole due to local damage.

3. 太空环境下的抗损能力

3. Damage resistance in the space environment

在太空环境中，飞行器面临微陨石、碎片撞击等不可预测的风险。对于传统一体式飞行器，任何关键部位的结构损伤都可能导致整机报废。而网状拼接结构将风险分散至各个独立单元：即使某个单元被撞击损坏，也仅仅是 "坏了一块积木"，整体框架和其余单元仍可正常工作。这一特性使系统在深空任务中具备远超传统飞行器的生存能力。

In the space environment, aircraft face unpredictable risks such as micrometeoroid and debris impacts. For conventional integrated aircraft, structural damage at any critical part can lead to scrapping the whole vehicle. The mesh-splicing structure disperses the risk across independent units: even if one unit is struck and damaged, it is merely “one block of the building blocks broken”, while

the overall frame and the remaining units keep working normally. This property gives the system survivability far beyond conventional aircraft in deep-space missions.

4. "坏一块，换一块" 的经济性

4. The economics of “break one, replace one”

网状结构的模块化设计带来了显著的全生命周期经济优势：

The modular design of the mesh structure brings notable full-lifecycle economic advantages:

无需整机报废

No need to scrap the whole vehicle

无需返厂大修

No need for factory overhauls

可进行在轨维修或地面快速更换

On-orbit repair or rapid ground replacement is possible

系统寿命大幅延长，单次任务摊销成本极低

System life is greatly extended, and the amortized cost per mission is extremely low

这种 "积木式" 的维修逻辑，使系统在全生命周期内保持了极高的可用性和极低的维护成本。

This “building-block” repair logic keeps the system highly available at extremely low maintenance cost over its full lifecycle.

7.2.5 多单元拼接的最小数量和推荐范围

7.2.5 Minimum Number and Recommended Range of Multi-Unit Splicing

最少数量：6 个单元

Minimum number: 6 units

6 个单元是形成稳定网状结构的最小数量。在此配置下：

Six units are the minimum needed to form a stable mesh structure. In this configuration:

可形成基本的蜂窝状或六边形布局

A basic honeycomb or hexagonal layout can be formed

具备初步的冗余容错能力（坏 1 个不影响整体，坏 2 个可通过调整补偿）

Preliminary redundancy fault tolerance exists (losing 1 unit does not affect the whole; losing 2 can be compensated by adjustment)

结构重量与载荷能力的比例在可接受范围内

The ratio of structural weight to lift capacity remains within an acceptable range

推荐范围：12-24 个单元

Recommended range: 12–24 units

在 12-24 个单元的范围内，系统的综合性能最优：

Within the range of 12–24 units the overall performance of the system is optimal:

12 个单元以上：冗余度显著提升，坏 2-3 个仍能正常工作

Above 12 units: redundancy rises markedly, and 2–3 failed units still allow normal operation

18 个单元左右：性价比最高，单位成本的载荷能力最优

Around 18 units: the best cost-effectiveness, with optimal payload capacity per unit cost

24 个单元以下：结构重量仍在可控范围内

Below 24 units: structural weight remains controllable

超过 24 个单元：边际效益递减

Beyond 24 units: diminishing marginal returns

超过 24 个单元后，每增加一个单元带来的额外载荷能力，会被结构重量增加带来的能耗上升所抵消。除非有特殊任务需求（如超重载荷），否则不建议超过 24 个单元。

Past 24 units, the extra payload capacity from each additional unit is offset by the increased energy consumption from added structural weight. Unless special

mission needs (such as super-heavy payloads) exist, exceeding 24 units is not recommended.

7.3 能量回收模块布局

7.3 Layout of the Energy-Recovery Modules

每个旋翼单元下方集成一个能量回收模块，包含：

Each rotor unit integrates one energy-recovery module below it, containing:

发电机单元：电机反向工作模式，将机械能转化为电能

Generator unit: the motor working in reverse mode, converting mechanical energy into electrical energy

整流电路：将交流电转换为直流电

Rectifier circuit: converting alternating current into direct current

电压调节器：稳定输出电压，适配电池充电要求

Voltage regulator: stabilizing the output voltage to match battery-charging requirements

能量管理控制器：实时监测发电功率，动态调整回收强度

Energy-management controller: monitoring generating power in real time and dynamically adjusting recovery intensity

布局原则：

Layout principles:

每个旋翼单元的能量回收模块独立工作，互不影响

The energy-recovery module of each rotor unit works independently without mutual interference

多个单元回收的电能通过汇流母线的形式汇集至主电池组

Energy recovered by multiple units is collected to the main battery pack through a common bus

汇流母线采用冗余设计，单一路径故障不影响整体回收

The common bus uses a redundant design; a single-path failure does not affect overall recovery

能量回收模块与动力电机共用同一套绕组和磁路，减少额外重量

The energy-recovery module shares the same windings and magnetic circuit with the power motor, reducing extra weight

电机双向工作的物理基础：

Physical basis of bidirectional motor operation:

电机既是电动机也是发电机，取决于电流方向和旋转方向的关系。

A motor is both an electric motor and a generator, depending on the relation between current direction and rotation direction.

正向模式：电流→电机旋转→产生推力（上升阶段）

Forward mode: current → motor rotation → thrust produced (ascent phase)

反向模式：气流推动桨叶旋转→电机反向旋转→产生电流（下降阶段）

Reverse mode: airflow drives blade rotation → motor turns backwards → current produced (descent phase)

这种 "双向工作" 能力不需要额外的机械切换，只需要电力电子电路的极性切换即可实现，响应时间在毫秒级。

This “bidirectional operation” capability requires no extra mechanical switching, only a polarity switch of the power-electronics circuit, with millisecond-level response time.

第 8 章 能量管理策略

Chapter 8 Energy-Management Strategy

本系统的能量管理涉及多个独立单元的同时充放电，需要一套智能的分配与调度策略。

The energy management of this system involves simultaneous charging and discharging of multiple independent units and requires an intelligent allocation and dispatch strategy.

8.1 上升阶段的能量分配策略

8.1 Energy-Allocation Strategy During Ascent

上升阶段，能量主要用于驱动旋翼单元产生升力。

During ascent, energy is mainly used to drive the rotor units to produce lift.

分配原则：

Allocation principles:

各单元根据其当前状态（温度、负载、健康度）分配功率

Power is allocated to each unit according to its current state (temperature, load, health)

状态良好、效率高的单元承担更多负载

Units in good condition and high efficiency take on more load

状态较差或温度过高的单元降低负载，作为保护

Units in poor condition or overheated are derated as protection

总功率需求由主控制器统一计算，分布式执行

Total power demand is computed centrally by the main controller and executed in a distributed manner

功率分配每 10 毫秒更新一次，适应飞行状态的实时变化

Power allocation is updated every 10 milliseconds to adapt to real-time changes of flight state

冗余切换逻辑：

Redundant switching logic:

某个单元失效时，其负载自动分配给邻近单元

When a unit fails, its load is automatically reassigned to neighboring units

切换时间小于 50 毫秒（人眼不可感知）

Switching time is less than 50 milliseconds (imperceptible to the human eye)

系统整体推力损失不超过 5%（当失效单元数量≤10% 时）

Overall thrust loss of the system does not exceed 5% (when the number of failed units is ≤10%)

负载均衡算法：

Load-balancing algorithm:

```
Plain Text IF 单元状态 = 健康 THEN 分配标准负载 IF 单元状态 = 过热 THEN 分配 80% 标准负载 IF 单元状态 = 故障 THEN 零分配，触发冗余切换 IF 单元状态 = 恢复中 THEN 分配 60% 标准负载，逐步恢复 Plain Text IF unit_state = healthy THEN assign standard load IF unit_state = overheated
```

THEN assign 80% standard load IF unit_state = failed THEN assign zero, trigger redundancy switching IF unit_state = recovering THEN assign 60% standard load, recover gradually

8.2 下降阶段的能量回收策略

8.2 Energy-Recovery Strategy During Descent

下降阶段，各旋翼单元进入发电模式。阻力面板展开程度和下降速度共同决定了回收功率。

During descent each rotor unit enters generating mode. The degree of drag-panel deployment and the descent speed jointly determine the recovery power.

回收策略：

Recovery strategy:

各单元独立发电，独立整流，独立调节

Each unit generates, rectifies and regulates independently

根据单元温度选择最佳发电负载（温度低可满负荷，温度高需降额）

The optimal generating load is chosen by unit temperature (full load when cool, derated when hot)

电池组充电状态（SOC）影响回收优先级：电池容量低时优先充电

The battery pack's state of charge (SOC) affects recovery priority: charging is prioritized when battery capacity is low

回收功率的动态平衡：当阻力面板展开到最大时，系统自动调节发电负载，使下降速度稳定在预设值

Dynamic balance of recovery power: when the drag panels are fully deployed, the system automatically adjusts the generating load so that the descent speed stays at the preset value

回收功率分配：

Allocation of recovered power:

总回收功率优先用于主电池组充电

Total recovered power is first used to charge the main battery pack

当主电池组接近满电时，将回收能量转移至辅助储能系统（超级电容或备用电池）

When the main pack nears full charge, recovered energy is diverted to auxiliary storage (supercapacitors or backup batteries)

紧急情况下，回收能量可直接供给其他单元（如姿态控制、加热系统）

In emergencies, recovered energy can directly supply other units (such as attitude control and heating systems)

发电负载与下降速度的关系：

Relation between generating load and descent speed:

发电负载越大，制动力越大，下降速度越慢

The larger the generating load, the greater the braking force and the slower the descent

下降速度越慢，单位时间内的回收功率反而降低（）

The slower the descent, the lower the recovery power per unit time ()

因此需要在 "尽可能慢" 和 "单位时间发电量最大化" 之间寻找平衡

Therefore a balance must be found between “as slow as possible” and “maximum generation per unit time”

最优下降速度：约 5-10 m/s（取决于具体配置）

Optimal descent speed: about 5–10 m/s (depending on the specific configuration)

8.3 应急能量储备

8.3 Emergency Energy Reserve

1/3 冗余储备原则应用于三个层面：

The 1/3 redundancy reserve principleapplies at three levels:

电池储备：

Battery reserve:

放电深度限制：不高于 80%（保留 20% 作为紧急储备）

Depth-of-discharge limit: no more than 80% (20% retained as emergency reserve)

紧急储备在任何情况下不被动用，除非系统判断存在坠毁风险

The emergency reserve is never touched unless the system judges that a crash risk exists

紧急储备的释放需要人工确认（地面控制台）或 AI 决策（极端情况下）

Release of the emergency reserve requires manual confirmation (ground console) or an AI decision (under extreme circumstances)

单元储备：

Unit reserve:

始终保留 1-2 个旋翼单元处于 "待机" 状态（低负载或零负载）

One or two rotor units are always kept in “standby” state (low or zero load)

当其他单元失效时，待机单元立即满负荷投入工作

When other units fail, the standby unit immediately joins at full load

待机单元轮流替换，避免长期待机导致的性能退化

Standby duty rotates among units to avoid performance degradation from prolonged idling

路径储备：

Path reserve:

上升路径规划中预留 "应急返回高度"—— 在达到该高度之前，下降回收的能量足以覆盖回程

The ascent path plan reserves an “emergency return altitude” — before reaching it, the energy recovered during descent is enough to cover the return

超出该高度后，系统进入 "单程模式"，只在目标点降落，不再返回

Beyond this altitude the system enters “one-way mode”, landing only at the destination and not returning

应急返回路径预先计算并存储在飞控系统中

Emergency return paths are pre-computed and stored in the flight-control system

8.4 1/3 冗余储备的边界条件与应急模式

8.4 Boundary Conditions of the 1/3 Redundancy Reserve and Emergency Mode

1/3 冗余的适用范围：

Scope of applicability of the 1/3 redundancy:

1/3 冗余（上升阶段能耗不超过总能量储备的 2/3）是在以下 "标准工况" 条件下计算得出的：

The 1/3 redundancy (ascent-phase energy consumption not exceeding 2/3 of the total energy reserve) is computed under the following “standard conditions”:

载荷重量在系统设计载重的 70%-100% 范围内

Payload weight within 70%–100% of the system's design load

目标高度不超过 50km（平流层中上部）

Target altitude not exceeding 50 km (upper-middle stratosphere)

天气条件良好（风速低于 15m/s，无强湍流）

Good weather conditions (wind speed below 15 m/s, no strong turbulence)

所有单元健康度在 90% 以上

Health of all units above 90%

极端工况下的冗余不足：

Insufficient redundancy under extreme conditions:

当以下多种极端工况叠加时，1/3 冗余可能不足以确保安全返回：

When several of the following extreme conditions overlap, the 1/3 redundancy may be insufficient to ensure a safe return:

载荷重量达到系统极限（100%-120%）

Payload weight reaches the system limit (100%–120%)

目标高度超过 50km（需消耗更多能量）

Target altitude exceeds 50 km (more energy must be consumed)

强风或湍流环境（需额外能耗维持姿态）

Strong wind or turbulence environment (extra energy needed to maintain attitude)

部分单元健康度下降（热损伤或轻微故障）

Degraded health of some units (thermal damage or minor faults)

应急模式：

Emergency mode:

在上述极端工况下，系统自动切换到 "应急模式":

Under the above extreme conditions the system automatically switches to “emergency mode”:

主动放弃部分回收效率：减少阻力面板展开程度，优先保证下降速度的安全范围

Deliberately give up part of the recovery efficiency: reduce drag-panel deployment, prioritizing the safe range of descent speed

能量优先用于安全返回：回收的电能不存入电池，而是直接供给姿态控制和关键单元

Energy first for safe return: recovered electricity is not stored in the battery but supplied directly to attitude control and critical units

简化下降路径：选择最短下降路径（牺牲发电量，换取更短留空时间）

Simplify the descent path: choose the shortest descent path (sacrificing generation for shorter loiter time)

应急功率分配：将可用能量全部集中于关键单元，非关键单元（如部分传感器）暂时关闭

Emergency power allocation: concentrate all available energy on critical units, temporarily shutting down non-critical ones (such as some sensors)

人工接管提示：如果情况仍不可控，系统请求地面控制台介入决策

Manual-takeover prompt: if the situation remains uncontrollable, the system requests ground-console intervention for decision

应急模式的触发条件:

Trigger conditions of emergency mode:

系统预测剩余能量不足以完成完整下降路径（基于当前高度、速度、载荷、天气的综合计算）

The system predicts the remaining energy is insufficient to complete the full descent path (based on integrated computation of current altitude, speed, payload and weather)

或主电池组 SOC（剩余电量）低于预设的安全阈值

Or the main battery pack's SOC (state of charge) falls below the preset safety threshold

应急模式的退出:

Exiting emergency mode:

当系统预测剩余能量足以完成安全返回时，自动切换回标准模式

When the system predicts the remaining energy suffices for a safe return, it automatically switches back to standard mode

或经地面控制台确认后，手动退出

Or exits manually after ground-console confirmation

8.5 多模式能量管理

8.5 Multi-Mode Energy Management

系统支持三种能量管理模式，根据任务需求自动切换：

The system supports three energy-management modes, switching automatically according to mission needs:

模式 Mode	适用场景 Applicable scenario	能量策略 Energy strategy
经济模式 Economy mode	短距升降、常规任务 Short-range lift, regular missions	优先回收、优先充电、能耗最低化 Recovery first, charging first, energy consumption minimized
标准模式 Standard mode	中等距离、常规载荷 Medium distance, regular payload	平衡能耗与回收，追求净能耗最小 Balance consumption and recovery, minimize net energy use
高性能模式 High-performance mode	重载、深空返回、极端任务 Heavy payload, deep-space return, extreme missions	能量全部用于任务完成，回收效率次之 All energy for mission completion, recovery efficiency secondary

模式切换逻辑：

Mode-switching logic:

系统根据任务输入参数（目标高度、载荷重量、预定返回时间）自动推荐最优模式

Based on mission input parameters (target altitude, payload weight, planned return time), the system automatically recommends the optimal mode

操作人员可手动覆盖（通过地面控制台或预设任务计划）

Operators can override manually (via the ground console or a preset mission plan)

模式切换在飞行过程中不受限，但切换时需保持飞行状态稳定

Mode switching is unrestricted during flight, but flight-state stability must be maintained during the switch

第 9 章 应用场景分析

Chapter 9 Application-Scenario Analysis

本系统的应用场景涵盖从近地运载到深空任务、从商业发射到应急救援的多个领域。

The application scenarios of this system span from near-Earth transport to deep-space missions, from commercial launches to emergency rescue.

9.1 独立太空往返飞行器

9.1 Independent Space Round-Trip Vehicle

这是本系统最核心、最颠覆性的应用场景：作为完全独立的太空往返飞行器—— 不依赖火箭、不依赖发射场、不依赖外部燃料补给。

This is the most core and most disruptive application scenario of the system: serving as a fully independent space round-trip vehicle — no dependence on rockets, launch sites, or external fuel resupply.

完整任务流程：

Complete mission flow:

地面起飞：系统从任意平坦地面垂直起飞（不需专用发射场）

Ground takeoff: the system lifts off vertically from any flat ground (no dedicated launch site needed)

上升阶段：旋翼单元提供升力，阻力面板收起，以最小阻力穿越大气层稠密区

Ascent phase: rotor units provide lift, drag panels stowed, crossing the dense atmospheric region at minimal drag

惯性滑出：到达平流层顶后，利用积累的惯性继续上升，进入近地轨道区域

Inertial run-out: after reaching the stratopause, accumulated inertia carries it further upward into the low-Earth-orbit region

太空航行：到达大气层外围后，系统以惯性维持轨道运动，几乎不消耗能量

Spaceflight: at the outer edge of the atmosphere, inertia maintains orbital motion with almost no energy consumption

任务执行：在近地轨道或亚轨道执行运载、观测、通信中继等任务

Mission execution: performs transport, observation, communication-relay and other tasks in low-Earth orbit or suborbital space

再入返回：完成预定任务后，通过调整姿态和推进方向进入预定再入窗口

Re-entry return: after completing the planned task, enters the designated re-entry window by adjusting attitude and thrust direction

叶子滑行：阻力面板展开，沿侧向路径缓慢进入大气层

Leaf gliding: drag panels deploy and the platform enters the atmosphere slowly along a lateral path

能量回收：下降过程中持续发电，回收能量回充电池组

Energy recovery: continuous generation during descent recharges the battery pack with recovered energy

降落复用：安全降落至指定地点，经检查后可再次使用

Landing and reuse: lands safely at the designated site and can be reused after inspection

与火箭发射的对比：

Comparison with rocket launches:

特性 Property	火箭发射 Rocket launch	本系统独立往返 Independent round trip by this system
发射场需求 Launch-site requirement	需要专用发射场 Dedicated launch site required	任何平地皆可 Any flat ground
燃料需求 Fuel requirement	几百吨化学燃料 Hundreds of tons of chemical fuel	只需电能 Only electrical energy
返程方式 Return method	燃料补给或回收失败 Fuel resupply or recovery failure	叶子滑行自然返回 Natural return by leaf gliding
复用率 Reuse rate	低（部分回收成本仍高） Low (partial recovery, cost still high)	高（数百次可用） High (hundreds of uses)
任务成本 Mission cost	数千万 - 数亿美元 Tens of millions - hundreds of millions of dollars	数百万 - 数千万美元 Millions - tens of millions of dollars
任务准备周期 Mission preparation cycle	数月 Months	数天至数周 Days to weeks

本系统为中等载荷（几百公斤 - 几吨）的近地轨道任务提供了 "火箭之外的新选择"。

This system offers “a new option beyond rockets” for medium-payload (hundreds of kilograms to a few tons) low-Earth-orbit missions.

9.2 星链（Starlink）类低轨卫星发射辅助

9.2 Launch Assistance for Starlink-Type LEO Constellations

当前低轨卫星星座发射面临的核心问题是：每次发射成本高、频率低、灵活性差。即使 SpaceX 的猎鹰 9 号将发射成本降低到了约每公斤几千美元，对于数千颗卫星的组网计划而言，总成本仍然极其高昂。

The core problem facing current LEO satellite-constellation launches is: each launch is costly, infrequent and inflexible. Even though SpaceX's Falcon 9 has cut launch cost to roughly a few thousand dollars per kilogram, for constellation plans of thousands of satellites the total cost remains extremely high.

本系统的解决方案：

The solution of this system:

将卫星集群部署在升降平台上，由平台送至平流层以上高度（约 30-50km）

Deploy the satellite cluster on the lift platform, which carries it to an altitude above the stratosphere (about 30–50 km)

平台以惯性滑行方式继续上升，接近大气层边缘

The platform continues rising by inertial coasting, approaching the edge of the atmosphere

卫星在此高度释放，启动自身小型推进器完成最终入轨

The satellites are released at this altitude and start their own small thrusters to complete final orbit insertion

与火箭发射的对比：

Comparison with rocket launches:

特性 Property	火箭发射 Rocket launch	本系统辅助发射 Launch assistance by this system
燃料消耗 Fuel consumption	极大（几万到几百吨） Extremely large (tens of thousands to hundreds of tons)	极小（电力 + 重力） Extremely small (electricity + gravity)

发射成本 Launch cost	极高（数千万 - 数亿美元） Extremely high (tens of millions - hundreds of millions of dollars)	较低（电费 + 维护） Lower (electricity + maintenance)
发射频率 Launch frequency	受限（场地、天气、燃料） Limited (site, weather, fuel)	灵活（电力补充即可） Flexible (electric recharge suffices)
环境影响 Environmental impact	大（燃料燃烧、噪音） Large (fuel combustion, noise)	小（电动运行、安静） Small (electric operation, quiet)
载荷重量限制 Payload weight limit	大载荷（几吨到几十吨） Heavy payload (a few tons to tens of tons)	中载荷（几百公斤到几吨） Medium payload (hundreds of kilograms to a few tons)
发射窗口 Launch window	受天气、发射场排期限制 Limited by weather and launch-site scheduling	可随时发射 Launch anytime

9.3 城际货物与人员运输

9.3 Inter-City Cargo and Passenger Transport

短途（几十 - 几百公里）：

Short range (tens to hundreds of kilometers):

垂直起降，无需跑道

Vertical takeoff and landing, no runway needed

电力驱动，噪音低，适合城市环境

Electrically driven, low noise, suitable for urban environments

可部署在楼顶、停车场等有限空间

Can be deployed on rooftops, parking lots and other limited spaces

速度适中，比直升机更安静、更经济

Moderate speed, quieter and more economical than helicopters

噪声水平：约 65-75 分贝（相比直升机的 90-100 分贝）

Noise level: about 65–75 dB (versus 90–100 dB for helicopters)

中途（几百 - 上千公里）：

Medium range (hundreds to over a thousand kilometers):

倾转旋翼模式，巡航飞行效率更高

Tiltrotor mode gives higher cruise efficiency

可在大气层内实现点对点运输

Point-to-point transport within the atmosphere is possible

能源成本远低于直升机（电 vs 航空燃油）

Energy cost far below helicopters (electricity vs. aviation fuel)

噪声低于直升机，噪音污染明显减少

Lower noise than helicopters, with clearly reduced noise pollution

经济性对比：

Economic comparison:

运输方式 Transport mode	每公里每吨能耗 Energy per km per ton	每公斤每公里成本 Cost per kg per km
直升机 Helicopter	高（燃油） High (fuel)	约 $2-5 about $2-5
电动垂直起降飞行器 Electric vertical takeoff and landing aircraft	中（电池） Medium (battery)	约 $1-3 about $1-3
本系统 This system	低（电力 + 回收） Low (electricity + recovery)	约 $0.3-1 about $0.3-1
卡车 / 火车 Truck / train	很低 Very low	约 $0.05-0.2 about $0.05-0.2

本系统在中短途运输中略贵于地面运输，但速度更快；在长途运输中，远低于直升机，接近电动垂直起降飞行器的经济性。

In short-to-medium transport this system is slightly more expensive than ground transport but faster; in long-range transport it is far below helicopters, approaching the economics of electric vertical takeoff and landing aircraft.

9.4 叶子滑行再入：深空返回的热防护突破

9.4 Leaf-Glide Re-Entry: A Thermal-Protection Breakthrough for Deep-Space Return

本系统最具潜在颠覆性的应用场景是：作为 "太空往返飞行器" 的组成部分。

The most potentially disruptive application of this system is: serving as a component of a “space round-trip vehicle”.

流程：

Flow:

火箭将载荷送至近地轨道或更远处

A rocket delivers the payload to low-Earth orbit or beyond

升降平台从地球表面起飞（电力驱动），升至平流层顶部

The lift platform takes off from Earth's surface (electrically driven) and climbs to the top of the stratosphere

平台作为太空往返飞行器的 "返回舱"，脱离火箭独立运行

The platform acts as the “return capsule” of the space vehicle, separating from the rocket to operate independently

完成太空任务后，返回地球大气层

After completing the space mission, it returns to Earth's atmosphere

叶子滑行模式：侧向滑行，缓慢减速，不依赖隔热盾

Leaf-glide mode: lateral gliding, slow deceleration, no dependence on a heat shield

磁阻力发电，回收下降能量

Magnetic-drag generation recovers the descent energy

安全降落，系统复用

Safe landing, system reuse

关键点：

Key points:

这不是 "可回收火箭" 的翻版 —— 完全不需要火箭燃料的返程

This is not a copy of “reusable rockets” — the return requires no rocket fuel at all

下降阶段不仅不消耗燃料，反而发电

The descent phase not only consumes no fuel but actually generates power

系统复用率极高（结构寿命可达数百次任务）

Extremely high reuse rate (structural life can reach hundreds of missions)

比任何现有可回收方案都更节能、更经济

More energy-saving and economical than any existing reusable scheme

9.5 其他潜在应用场景

9.5 Other Potential Application Scenarios

场景 Scenario	描述 Description	本系统优势 Advantage of this system
应急救援 Emergency rescue	山体滑坡、洪水、地震区域的物资投送 Material delivery to landslide, flood and earthquake areas	垂直起降、无需跑道、可在极端环境部署 Vertical takeoff and landing, no runway, deployable in extreme environments
高空基站 High-altitude base station	搭载通信设备，在平流层提供宽带覆盖 Carrying communication equipment to provide broadband coverage from the stratosphere	长时间悬停、太阳能充电、可多平台协同组网 Long loiter, solar charging, multi-platform cooperative networking
科学研究 Scientific research	大气层采样、高空天文观测、气候研究 Atmospheric sampling, high-altitude astronomical observation, climate research	可精细调节高度和位置、可停留较长时间 Fine adjustment of altitude and position, long loiter possible
军事应用 Military applications	战场侦查、物资补给、通信中继 Battlefield reconnaissance, supply delivery, communication relay	低可探测性（电动）、高冗余生存能力 Low detectability (electric), high-redundancy survivability
观光旅游 Sightseeing and tourism	高空观光旅游、亚轨道体验 High-altitude sightseeing, suborbital experience	安静、平稳、可重复使用 Quiet, smooth, reusable

9.6 应用场景优先级排序

9.6 Priority Ranking of Application Scenarios

基于技术成熟度、市场需求和竞争格局，建议按以下优先级推进应用：

Based on technology maturity, market demand and competitive landscape, applications are recommended to advance in the following priority:

	应用场景 Application scenario	理由 Rationale

优先级 Priority		
1 1	应急救援与物资投送 Emergency rescue and material delivery	技术门槛相对低，市场需求迫切 Relatively low technical threshold, urgent market demand
2 2	高空基站与通信中继 High-altitude base station and communication relay	现有技术存在空白，竞争少 Gap in existing technology, little competition
3 3	星链类卫星发射辅助 Starlink-type satellite launch assistance	市场巨大，但需要更多验证 Huge market, but more validation needed
4 4	城际货物与人员运输 Inter-city cargo and passenger transport	需要空域管理和法规配套 Needs airspace management and supporting regulations
5 5	独立太空往返飞行器 Independent space round-trip vehicle	技术最为复杂，但最具颠覆性 Technically most complex, yet most disruptive

第 10 章 与现有方案的对比

Chapter 10 Comparison with Existing Schemes

本章从多个维度对比本系统与现有运载方案的优劣。

This chapter compares the strengths and weaknesses of this system against existing transport schemes along multiple dimensions.

10.1 传统火箭

10.1 Traditional Rockets

特性 Property	火箭 Rocket	本系统 This system
推力来源 Thrust source	化学燃料燃烧 Chemical fuel combustion	电力驱动（电池 + 回收） Electric drive (battery + recovery)
发射成本 Launch cost	极高（数千万 - 数亿美元 / 次） Extremely high (tens of millions - hundreds of millions of dollars per launch)	较低（数百万 - 数千万美元 / 次） Lower (millions - tens of millions of dollars per launch)
复用率 Reuse rate	低（需回收或部分回收） Low (requires recovery or partial recovery)	高（结构复用数百次） High (structure reused hundreds of times)

可维护性 Maintainability	差（返厂大修） Poor (factory overhaul)	好（模块化更换） Good (modular replacement)
安全冗余 Safety redundancy	低（单点故障风险高） Low (high single-point-failure risk)	高（网状冗余、容错设计） High (mesh redundancy, fault-tolerant design)
环境友好度 Environmental friendliness	差（燃料燃烧污染、噪音巨大） Poor (fuel-combustion pollution, huge noise)	良（电动、低噪音） Good (electric, low noise)
发射场需求 Launch-site requirement	需要专用发射场 Dedicated launch site required	任何平地皆可 Any flat ground
发射准备周期 Launch preparation cycle	数月 Months	数天至数周 Days to weeks
载荷重量范围 Payload weight range	大（几吨 - 几十吨） Large (a few tons - tens of tons)	中（几百公斤 - 几吨） Medium (hundreds of kg - a few tons)

10.2 直升机 / 倾转旋翼机

10.2 Helicopters / Tiltrotor Aircraft

特性 Property	直升机 Helicopter	本系统 This system
升限 Service ceiling	约 6-8km About 6-8 km	可突破大气层边缘（约 100km） Can reach the edge of the atmosphere (about 100 km)
速度 Speed	约 200-300km/h About 200-300 km/h	大气层内中等速度，大气层外惯性滑行 Medium speed within the atmosphere, inertial coasting outside
航程 Range	几百公里（需加油） Hundreds of kilometers (refueling needed)	可长时间飞行（能量回收） Long-duration flight possible (energy recovery)
能耗 Energy consumption	高（航空燃油） High (aviation fuel)	低（电力 + 回收） Low (electricity + recovery)
容错能力 Fault tolerance	低（单旋翼失效即坠落） Low (single-rotor failure means falling)	高（多单元冗余） High (multi-unit redundancy)
噪音 Noise		较低（65-75 分贝） Lower (65-75 dB)

	极高（90-100 分贝） Extremely high (90-100 dB)	
维护成本 Maintenance cost	高（复杂机械结构） High (complex mechanical structure)	低（模块化更换） Low (modular replacement)
垂直起降能力 Vertical takeoff-and-landing capability	有 Yes	有 Yes
跑道需求 Runway requirement	无需 Not needed	无需 Not needed

10.3 大型无人机

10.3 Large Drones

特性 Property	无人机 Drone	本系统 This system
载荷能力 Payload capacity	几十 - 几百公斤 Tens - hundreds of kilograms	几百公斤 - 几吨 Hundreds of kilograms - a few tons
升限 Service ceiling	约 10-15km About 10-15 km	可突破大气层（约 100km） Can break through the atmosphere (about 100 km)
续航时间 Endurance	数小时（电池限制） Hours (battery-limited)	较长（能量回收延长） Longer (extended by energy recovery)
可扩展性 Scalability	有限（单机设计） Limited (single-aircraft design)	强（网状拼接、可叠加单元） Strong (mesh splicing, units stackable)
冗余容错 Redundancy fault tolerance	有限（部分型号有冗余） Limited (some models have redundancy)	高（全面冗余） High (full redundancy)
噪音 Noise	中 Medium	中低 Medium-low
维护成本 Maintenance cost	中 Medium	低（模块化） Low (modular)

10.4 综合对比结论

10.4 Overall Comparison Conclusion

本系统不是任何现有运载方案的 "替代品"，而是填补了现有方案之间的空白：

This system is not a “replacement” for any existing transport scheme; it fills the gaps between existing schemes:

比火箭便宜，比直升机飞得高

Cheaper than rockets, flying higher than helicopters

比无人机运力大，比火箭可复用性好

Greater lift capacity than drones, better reusability than rockets

不依赖燃料，不依赖跑道，不依赖发射场

No dependence on fuel, no dependence on runways, no dependence on launch sites

在中等高度（0-100km）、中等载荷（几百公斤 - 几吨）的任务区间具有独特的成本优势

It holds a unique cost advantage in the mission band of medium altitude (0–100 km) and medium payload (hundreds of kilograms to a few tons)

本系统最适用的任务特征：

Mission characteristics best suited to this system:

中等载荷：几百公斤到几吨（非几十吨级别）

Medium payload: hundreds of kilograms to a few tons (not tens-of-tons class)

中等高度：0-100km（非深空距离）

Medium altitude: 0–100 km (not deep-space distances)

需要垂直起降（无跑道）

Vertical takeoff and landing required (no runway)

需要多次复用（经济性要求高）

Multiple reuses required (high economic requirements)

需要快速响应（发射窗口灵活）

Rapid response required (flexible launch windows)

本系统不适用于以下任务：

This system is unsuitable for the following missions:

超重载荷（超过 10 吨）

Super-heavy payloads (above 10 tons)

超远距离（超过近地轨道）

Ultra-long distances (beyond low-Earth orbit)

需要极高速度（如拦截导弹）

Extremely high speeds required (such as missile interception)

第 11 章 局限性与未来方向

Chapter 11 Limitations and Future Directions

11.1 当前的理论局限

11.1 Current Theoretical Limitations

局限一：高海拔电力供应不足

Limitation 1: insufficient power supply at high altitude

在平流层以上高度（>50km），空气密度极低，旋翼单元的推力效率大幅下降。本系统在高海拔段的爬升能力依赖于惯性积累，而非持续推力。对于需要 "持续爬升到超高高度" 的任务，仍需辅助推进手段。

Above the stratosphere (>50 km) air density is extremely low and the thrust efficiency of rotor units falls sharply. The platform's high-altitude climbing capability relies on accumulated inertia rather than sustained thrust. For missions requiring “continuous climb to ultra-high altitudes”, auxiliary propulsion is still needed.

应对思路： 在更高海拔段（>50km）采用小型电推或等离子体推进作为辅助，或采用 "弹弓式" 轨道 —— 在地面附近加速，利用惯性完成剩余的爬升。

Coping idea: in the higher-altitude band (>50 km), adopt small electric propulsion or plasma propulsion as an aid, or use a “slingshot” trajectory — accelerate near the ground and finish the remaining climb by inertia.

局限二：水平航程有限

Limitation 2: limited horizontal range

在纯垂直起降模式下，本系统的水平移动能力有限。虽然倾转旋翼单元支持水平巡航，但在网状拼接结构下，整体水平航程仍受限于电池容量和空气动力学效率。

In pure vertical-takeoff mode, the platform's horizontal mobility is limited. Although tiltrotor units support horizontal cruise, under the mesh-splicing structure the overall horizontal range remains constrained by battery capacity and aerodynamic efficiency.

应对思路： 采用可折叠机翼（水平飞行时展开），在巡航阶段将多旋翼结构转换为固定翼模式，提高升阻比。或采用 "接力" 模式 —— 多个平台接力运输，中途换手。

Coping idea: adopt foldable wings (deployed during horizontal flight), converting the multi-rotor structure to fixed-wing mode during cruise to raise the lift-to-drag ratio; or use a “relay” mode — multiple platforms relay the transport, handing over en route.

局限三：极端天气适应性

Limitation 3: extreme-weather adaptability

强风、湍流、极端温度条件下的性能表现尚待验证。虽然冗余结构能在一定程度上应对不利环境，但在台风、暴雪等极端天气下的飞行能力可能存在显著限制。

Performance under strong wind, turbulence and extreme temperature has yet to be verified. Although the redundant structure can cope with adverse environments to some extent, flight capability in typhoons, blizzards and other extreme weather may be significantly limited.

应对思路： 限制任务气象条件（风速、温度范围），待系统成熟后逐步放宽。或开发 "抗风模式"—— 调整单元布局，降低风阻。

Coping idea: restrict mission weather conditions (wind-speed and temperature ranges), relaxing them gradually as the system matures; or develop an “anti-wind mode” that adjusts unit layout to reduce wind drag.

局限四：控制算法复杂度

Limitation 4: control-algorithm complexity

数十个独立单元协同工作，对控制算法提出了较高要求。随着单元数量增加，控制系统的复杂度呈指数级增长。现有飞控算法是否能够支持大规模网状拼接结构的稳定控制，尚需实验验证。

Dozens of independent units working in coordination place high demands on control algorithms. As the number of units grows, control-system complexity

increases exponentially. Whether existing flight-control algorithms can support stable control of a large-scale mesh-spliced structure remains to be verified experimentally.

应对思路： 采用 "主从式" 控制架构 —— 主控制器负责高层决策（路径规划、能量管理），各单元独立执行低层控制（姿态稳定、速度调节）。主控制器不干预每个单元的细节，减少计算负担。

Coping idea: adopt a “master-slave” control architecture — the master controller handles high-level decisions (path planning, energy management), while each unit independently executes low-level control (attitude stabilization, speed regulation). The master controller does not intervene in the details of each unit, reducing the computational burden.

局限五：载荷重量与能耗的正反馈限制

Limitation 5: the positive-feedback limit of payload weight and energy consumption

虽然 "重力越大，发电越多" 是优势，但也存在一个边界：当载荷重量过大时，上升阶段的能耗增长快于下降阶段的发电增长，导致净能耗为负。这个边界值需要通过实验确定。

Although “the greater the gravity, the more power generated” is an advantage, a boundary exists: when payload weight is too large, ascent energy consumption grows faster than descent generation, making net energy negative. This boundary value must be determined experimentally.

应对思路： 给出推荐载荷范围（如 "推荐载重为系统设计载重的 60%-100%"），超出该范围则不建议使用。

Coping idea: provide a recommended payload range (e.g., “recommended load is 60%–100% of the system's design load”); use beyond this range is not recommended.

11.2 未来研究方向

11.2 Future Research Directions

方向一：高海拔辅助推进

Direction 1: high-altitude auxiliary propulsion

研究在高海拔段（30-100km）辅助推进方案，如小型电推、等离子体推进等，增强高海拔段的爬升能力和任务灵活性。

Study auxiliary propulsion schemes for the high-altitude band (30–100 km), such as small electric propulsion and plasma propulsion, to strengthen high-altitude climb capability and mission flexibility.

方向二：太阳能 - 电力混合能源系统

Direction 2: solar-electric hybrid energy system

在阻力面板表面集成薄膜太阳能电池，在白天飞行时持续补充电能。太阳能与阻力回收形成双重能源来源，进一步延长续航时间和任务半径。

Integrate thin-film solar cells on the drag-panel surfaces to continuously replenish electrical energy during daytime flight. Solar power and drag recovery form dual energy sources, further extending endurance and mission radius.

方向三：智能分布式控制算法

Direction 3: intelligent distributed control algorithms

开发基于神经网络的分布式控制算法，使网状拼接结构具备 "自组织" 能力 —— 部分单元失效时，其余单元自动重新组织升力分布，无需主控制器干预。

Develop neural-network-based distributed control algorithms to give the mesh-spliced structure “self-organizing” capability — when some units fail, the rest automatically reorganize the lift distribution without master-controller intervention.

方向四：规模化实验验证

Direction 4: scale-up experimental verification

从 6 单元、12 单元到 18 单元、36 单元，逐步验证系统的可扩展性和性能极限。建立不同规模下的性能数据库，为工程应用提供设计依据。

Progressively verify scalability and performance limits from 6 units and 12 units to 18 and 36 units, building a performance database at different scales to provide design bases for engineering application.

方向五：成本 - 收益模型分析

Direction 5: cost-benefit model analysis

建立包含制造成本、运行成本、维护成本、寿命周期在内的完整经济模型，量化本系统相比火箭、直升机、无人机的经济优势，为其商业化提供数据支撑。

Build a complete economic model covering manufacturing cost, operating cost, maintenance cost and life cycle, quantifying this system's economic advantage over rockets, helicopters and drones to provide data support for its commercialization.

方向六：极端环境适应性研究

Direction 6: extreme-environment adaptability research

通过风洞实验和数值模拟，研究系统在强风、湍流、极端温度条件下的性能表现，建立"安全飞行包线"，为实际运营提供依据。

Through wind-tunnel tests and numerical simulation, study system performance under strong wind, turbulence and extreme temperature, establishing a “safe flight envelope” to support actual operations.

第 12 章 实验验证

Chapter 12 Experimental Verification

12.1 最经济的实验验证路径

12.1 The Most Economical Experimental Verification Path

本系统的验证不需要一步到位。推荐以下分阶段验证路径：

Verification of this system need not be done in one step. The following staged verification path is recommended:

第一阶段：3-4 个单元的小型验证平台

Stage 1: small verification platform of 3–4 units

目标：验证 "上升耗电、下降发电" 的基本假设是否成立

Goal: verify whether the basic premise “consume energy on the way up, generate power on the way down” holds

用 3-4 个旋翼单元拼接成最小网状结构

Splice 3–4 rotor units into the minimal mesh structure

在 10-50 米高度进行起降测试

Conduct takeoff-and-landing tests at 10–50 m altitude

测量上升阶段耗电量和下降阶段发电量

Measure ascent-phase energy consumption and descent-phase power generation

如果发电量 > 耗电量的 40%（初步目标），说明原理成立

If generation exceeds 40% of consumption (preliminary target), the principle is established

成本：约 $5,000-20,000（取决于旋翼单元的选择）

Cost: about $5,000–20,000 (depending on the choice of rotor units)

第二阶段：6-8 个单元的扩展验证平台

Stage 2: extended verification platform of 6–8 units

目标：验证容错能力、冗余切换、阻力面板效果

Goal: verify fault tolerance, redundancy switching and drag-panel effectiveness

在 6-8 个单元规模下重复测试

Repeat the tests at the 6–8-unit scale

模拟部分单元失效，观察系统响应

Simulate partial unit failures and observe the system response

测试阻力面板展开 / 收起对发电量的影响

Test the effect of drag-panel deployment/retraction on power generation

成本：约 $20,000-50,000

Cost: about $20,000–50,000

第三阶段：12-18 个单元的功能验证平台

Stage 3: functional verification platform of 12–18 units

目标：验证大尺寸系统的性能和稳定性

Goal: verify the performance and stability of the large-scale system

在目标载荷下进行起降测试（几百公斤级别）

Conduct takeoff-and-landing tests at the target payload (hundreds-of-kilograms class)

测试在不同高度、不同天气条件下的性能

Test performance under different altitudes and weather conditions

收集完整性能数据，优化参数配置

Collect complete performance data and optimize parameter configuration

成本：约 $100,000-300,000

Cost: about $100,000–300,000

第四阶段：全规模原型机

Stage 4: full-scale prototype

目标：验证最终设计方案的完整功能

Goal: verify the complete function of the final design

在目标载荷和高度下完成全程测试

Finish full-profile tests at the target payload and altitude

进行多次起降循环，验证复用性和耐久性

Run multiple takeoff-and-landing cycles to verify reusability and durability

成本：约 $500,000-2,000,000

Cost: about $500,000–2,000,000

验证时间估算：

Verification time estimate:

第一阶段：1-2 个月

Stage 1: 1–2 months

第二阶段：2-3 个月

Stage 2: 2–3 months

第三阶段：3-6 个月

Stage 3: 3–6 months

第四阶段：6-12 个月

Stage 4: 6–12 months

总计：约 12-24 个月

Total: about 12–24 months

12.2 单元测试

12.2 Unit Tests

单元测试验证单个组件的基本性能：

Unit tests verify the basic performance of individual components:

测试单个旋翼单元（倾转旋翼和涵道旋翼）的推力和能耗

Test the thrust and energy consumption of a single rotor unit (tiltrotor and ducted rotor)

测试阻力面板的展开 / 收起响应时间（目标：<0.5 秒）

Test the deployment/retraction response time of the drag panels (target: <0.5 s)

测试发电模块在不同风速下的输出功率（目标：在 8-15m/s 风速下效率 > 70%）

Test the output power of the generating module at different wind speeds (target: efficiency >70% at 8–15 m/s wind speed)

测试电池组在不同放电深度下的循环寿命

Test the cycle life of the battery pack at different depths of discharge

12.3 系统测试

12.3 System Tests

系统测试验证多个单元协同工作时的整体性能：

System tests verify overall performance when multiple units work in coordination:

在 3-4 个单元的小型平台上测试基本升降功能

Test basic lift functions on the small 3–4-unit platform

在 6-8 个单元的中型平台上测试上升阶段能耗和下降阶段发电量

Test ascent-phase energy consumption and descent-phase generation on the medium 6–8-unit platform

在 12-18 个单元的大型平台上测试部分单元失效时的容错能力

Test fault tolerance under partial unit failure on the large 12–18-unit platform

测试不同阻力面板展开程度对发电量的影响

Test the effect of different drag-panel deployment degrees on power generation

12.4 关键实验指标

12.4 Key Experimental Metrics

测试项目 Test item	预期结果 Expected result	验证标准 Validation criterion
上升阶段空气阻力减少率 Air-drag reduction rate during ascent	40%-60% 40%-60%	与传统设计对比 Compared with conventional designs

下降阶段能量回收率 Energy-recovery rate during descent	80%-90% 80%-90%	与理论计算对比 Compared with theoretical calculation
典型工况下净能量 Net energy under typical conditions	正值（发电＞消耗） Positive (generation > consumption)	连续 10 次起降平均 Average over 10 consecutive takeoff-landing cycles
失效单元容错能力 Fault tolerance to failed units	可容忍 10%-20% 单元失效 Tolerates 10%-20% unit failure	主动切断单元测试 Active unit-cutoff test
阻力面板响应时间 Drag-panel response time	<0.5 秒 <0.5 s	高速摄像机测量 Measured with a high-speed camera
系统整体可靠性 Overall system reliability	>99%（单次任务） >99% (single mission)	统计测试数据 Statistical test data
结构疲劳寿命 Structural fatigue life	>500 次起降 >500 takeoffs and landings	加速疲劳测试 Accelerated fatigue testing

12.5 实验数据记录与分析

12.5 Recording and Analysis of Experimental Data

应记录的数据项：

Data items to be recorded:

上升阶段：初始高度、目标高度、总耗时、总耗电量（Wh）、各单元功率分配、环境温湿度、风速

Ascent phase: initial altitude, target altitude, total duration, total energy consumed (Wh), power allocation of each unit, ambient temperature and humidity, wind speed

下降阶段：初始高度、下降速度、总耗时、总发电量（Wh）、各单元发电功率、电池 SOC 变化

Descent phase: initial altitude, descent speed, total duration, total generation (Wh), generating power of each unit, battery SOC changes

环境参数：温度、湿度、风速、风向、气压

Environmental parameters: temperature, humidity, wind speed, wind direction, air pressure

系统状态：各单元温度、振动数据（加速度计）、姿态角

System state: temperature of each unit, vibration data (accelerometer), attitude angles

数据分析方法：

Data-analysis methods:

计算净能量 = 发电量 - 耗电量（正值为正收益）

Compute net energy = generation − consumption (a positive value means net gain)

计算能量回收率 = 发电量 / 下降重力势能

Compute energy-recovery ratio = generation / descent gravitational potential energy

绘制 "载荷重量 vs 净能量" 曲线，找到最优载荷范围

Plot the “payload weight vs. net energy” curve to find the optimal payload range

绘制 "下降速度 vs 发电功率" 曲线，验证 关系

Plot the “descent speed vs. generating power” curve to verify the relation

统计单元失效频率，验证系统可靠性指标

Count unit-failure frequency to verify system reliability metrics

12.6 实验结果的工程迭代逻辑

12.6 Engineering-Iteration Logic of Experimental Results

实验的目的不是一次验证，而是发现需要调整的地方，然后调整，再验证。

The purpose of experiments is not one-shot verification but discovering what needs adjustment, adjusting, and verifying again.

迭代循环：

Iteration loop:

设计实验 → 2. 获取数据 → 3. 识别问题 → 4. 调整设计 → 5. 重新实验

Design experiment → 2. Obtain data → 3. Identify problems → 4. Adjust design → 5. Re-experiment

常见问题及调整方向：

Common problems and adjustment directions:

实验发现 Experimental finding	可能的调整方向 Possible adjustment direction
上升耗电太大 Ascent consumes too much energy	增加旋翼单元数量、减重、优化上升速度曲线 Add rotor units, reduce weight, optimize the ascent speed profile
下降发电太少 Descent generates too little power	增大阻力面板面积、优化下降速度、调整发电负载 Enlarge drag-panel area, optimize descent speed, adjust generating load
单元频繁失效 Frequent unit failures	加强散热、改进控制算法、降低负载 Improve heat dissipation, refine control algorithms, reduce load
结构振动过大 Excessive structural vibration	增加阻尼、优化框架刚度、调整旋翼转速匹配 Add damping, optimize frame stiffness, match rotor speeds
净能量为负 Net energy negative	降低目标高度、减少载荷、增加阻力面板面积 Lower the target altitude, reduce payload, enlarge drag panels

工程优化的本质： 本文提出的架构是固定的，但参数（单元数量、面板面积、电池容量、电机功率）都是可调的。实验的价值在于：用真实数据告诉你 “这个参数应该调到多少”。

The essence of engineering optimization: The architecture proposed in this paper is fixed, but the parameters (unit count, panel area, battery capacity, motor power) are all adjustable. The value of experiments lies in: real data tells you “what this parameter should be tuned to”.

第 13 章 总结

Chapter 13 Summary

13.1 核心贡献回顾

13.1 Review of Core Contributions

提出了空气阻力双向收割的运载范式。 将空气阻力从 "必须克服的敌人" 转变为 "可以被利用的资源"。

Proposed the transport paradigm of bidirectional air-drag harvesting. Air drag is turned from “an enemy that must be overcome” into “a resource that can be used”.

设计了多旋翼网状拼接结构。 通过多个独立单元的拼接，实现了高冗余容错能力和 "积木式" 可维护性。

Designed the multi-rotor mesh-splicing structure. Through the splicing of multiple independent units, high redundancy fault tolerance and “building-block” maintainability are achieved.

证实了能量闭环的可行性。 上升阶段省能 + 下降阶段发电 ≥ 上升阶段消耗，使系统具备自我维持能力。

Demonstrated the feasibility of the closed-loop energy cycle. Ascent energy saved + descent generation ≥ ascent consumption, giving the system self-sustaining capability.

提出了叶子滑行再入策略。 以侧向滑行路径取代垂直坠落，无需隔热盾即可安全再入大气层。

Proposed the leaf-glide re-entry strategy. A lateral gliding path replaces vertical falling, allowing safe atmospheric re-entry without a heat shield.

论证了重力载荷与发电量的正反馈关系。 颠覆了 "重量是负担" 的传统认知，建立了 "重量是能量来源" 的新视角。

Established the positive-feedback relation between gravity load and power generation. Overturning the traditional view that “weight is a burden”, it builds the new perspective that “weight is an energy source”.

确立了 2/3 能量红线的工程约束。 将 "能回来" 置于 "能上去" 之上，确保系统在所有工况下都可重复使用。

Set the 2/3 energy red-line engineering constraint. Placing “being able to come back” above “being able to go up”, ensuring the system is reusable under all operating conditions.

13.2 技术成熟度评估

13.2 Technology Readiness Assessment

基于当前的理论分析和初步估算，本系统的技术成熟度（TRL）约为 3-4 级（概念验证阶段）。

Based on the current theoretical analysis and preliminary estimates, the technology readiness level (TRL) of this system is about 3–4 (concept-validation stage).

TRL等级 TRL level	描述 Description	本系统状态 Status of this system

1 1	基本原理观察到 Basic principles observed	✅ 已完成 ✅ Completed
2 2	概念设计 Conceptual design	✅ 已完成 ✅ Completed
3 3	关键功能实验验证 Experimental validation of key functions	⏳ 待完成（小型验证平台） ⏳ Pending (small verification platform)
4 4	实验室环境原型验证 Prototype validation in laboratory environment	⏳ 待完成（中型验证平台） ⏳ Pending (medium verification platform)
5 5	相关环境原型验证 Prototype validation in relevant environment	⏳ 待完成（大型验证平台） ⏳ Pending (large verification platform)
6 6	工程原型演示 Engineering prototype demonstration	⏳ 待完成 ⏳ Pending
7 7	飞行演示 Flight demonstration	⏳ 待完成 ⏳ Pending
8 8	实际系统完成 Actual system completed	⏳ 待完成 ⏳ Pending
9 9	实际任务验证 Mission-proven in actual tasks	⏳ 待完成 ⏳ Pending

13.3 最终定位

13.3 Final Positioning

本系统不是对现有飞行器的改进，而是一种新的运载范式。它不依赖燃料，不依赖跑道，不依赖发射场，不依赖单一关键部件，不依赖外部供能。

This system is not an improvement of existing aircraft but a new transport paradigm. It depends on no fuel, no runway, no launch site, no single critical component, and no external energy supply.

它把大气层当作 "跳板" 和 "能量交换介质"，把重力当作 "能量来源"，把空气阻力当作 "资源" 而非 "障碍"。

It treats the atmosphere as a “springboard” and “energy-exchange medium”, gravity as an “energy source”, and air drag as a “resource” rather than an “obstacle”.

它既是一个升降平台，也是一个可重复使用的运载系统，在特定任务区间（中等高度、中等载荷、高频起降）具有独特的成本和效率优势。

It is both a lift platform and a reusable transport system, holding unique cost and efficiency advantages in a specific mission band (medium altitude, medium payload, high-frequency takeoff and landing).

13.4 结语

13.4 Conclusion

本研究提出了一种不依赖传统燃料推进的升降平台飞行器系统，通过空气阻力双向收割和多旋翼网状拼接结构的设计，实现了能量的闭环循环和系统的高冗余容错。

This study presents a lift-platform aircraft system that does not rely on traditional fuel propulsion, achieving a closed-loop energy cycle and high system redundancy fault tolerance through the design of bidirectional air-drag harvesting and a multi-rotor mesh-splicing structure.

该系统在理论上具备上升省力、下降发电、部分单元失效不影响整体、叶子滑行安全再入等核心能力，可在星链发射辅助、城际运输、深空往返等多个场景中发挥作用。

In theory the system features energy saving on ascent, power generation on descent, immunity to partial unit failures, and safe leaf-glide re-entry, and can serve in scenarios such as Starlink launch assistance, inter-city transport, and deep-space round trips.

后续研究将通过规模化实验验证其实际性能，并通过工程迭代优化参数配置，推动该系统从概念设计走向工程应用。

Follow-up research will verify actual performance through scale-up experiments and optimize parameter configuration through engineering iteration, moving the system from conceptual design toward engineering application.

本系统的最终目标是：

The ultimate goal of this system is:

让飞行不再依赖燃料的燃烧，而是依赖对重力和空气阻力的智能利用。

To make flight no longer dependent on burning fuel, but on the intelligent use of gravity and air drag.

让运载不再依赖发射场的设施，而是依赖模块化拼接结构的自我构建能力。

To make transport no longer dependent on launch-site facilities, but on the self-assembly capability of modular splicing structures.

让往返太空不再是一次性消耗品，而是一种可重复使用、可自我维持的常态交通方式。

To make space round trips no longer disposable consumables, but a reusable, self-sustaining, routine mode of transportation.

它是一个升降平台。

It is a lift platform.

但它的意义，远远超过一个"平台"。

But its significance goes far beyond a “platform”.